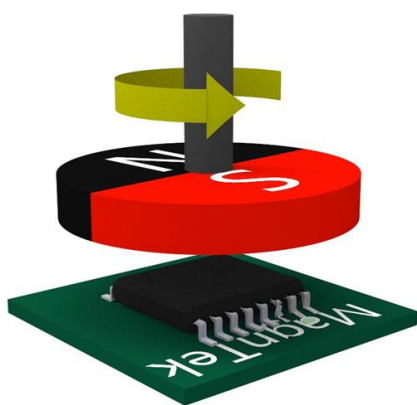


21-Bit High Accuracy Magnetic Angle Encoder IC

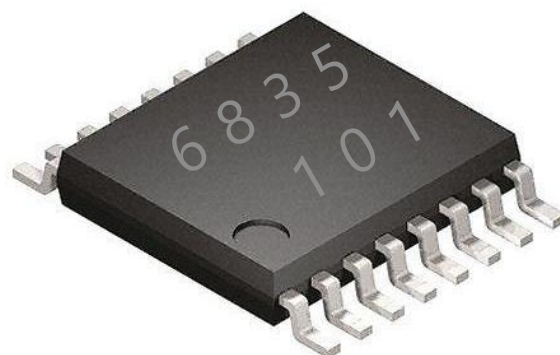
Features and Benefits

- Based on advanced AMR Technology with 0~360° Full Range Angle Sensing
- 21-bit Core Resolution with Maximum Rotation Speed Up to 120,000 RPM
- Output Propagation Delay <10 us
- User Auto-Calibration and Distortion Compensation with Target INL < $\pm 0.07^\circ$
- -40~125°C Operating Temperature Range
- Independent Output Interface: ABZ, UVW, PWM and SPI
- Incremental ABZ Resolution 1~16,384 Pulses per Revolution User Programmable
- UVW Output Resolution 1~16 Pole-Pairs per Revolution User Programmable
- 3.3~5.0V Programmable EEPROM



Applications

- Absolute Angle Position Sensor
- BLDC Motor Control
- Servo Motor Control
- Stepping Motor Control
- Optical Encoder Replacement



General Description

MagnTek's rotary position sensor MT6835 is an IC based on advanced AMR technology. A rotating magnetic field in the x-y sensor plane delivers two sinusoidal output signals which indicating the angle (α) between the sensor and the magnetic field direction.

The sensor is only sensitive to the magnetic field direction in x-y plane as the sensing element output is specially designed to be independent from the magnetic field strength. This allows the device to be less sensitive to magnet variations, stray magnetic fields, air gap changes and off-axis misalignment.

The incremental ABZ output mode is available in this sensor series, making the chip suitable to replace various optical encoders. The maximum resolution is 16,384 pulses or 65,536 steps per revolution

A 4-Wire SPI interface allows a host microcontroller to read out the 21-bit absolute angle position data from MT6835. The absolute angle position is also provided as a 12-bit PWM output.

Table of Contents

Features and Benefits.....	1
Applications.....	1
General Description.....	1
1 Pin Configuration.....	3
2 Function Diagram.....	4
3 Absolute Maximum Ratings.....	5
4 Operating Conditions.....	5
5 Electrical Characteristics.....	6
6 Magnetic Input Specifications.....	9
7 Output Mode.....	10
7.1 I/O Pin Configuration.....	10
7.2 Reference Circuit.....	11
7.3 Quadrature A,B and Zero-Position Output Signal (ABZ Mode).....	12
7.4 UVW Output Mode.....	17
7.5 Pulse Width Modulation (PWM) Output Mode.....	18
7.6 SPI Interface.....	19
7.6.1 SPI Reference Circuit.....	19
7.6.2 SPI Timing Diagram.....	20
7.6.3 SPI Protocol.....	21
7.6.4 SPI Read One Byte Register.....	22
7.6.5 SPI Write One Byte Register.....	22
7.6.6 SPI Program EEPROM.....	23
7.6.7 SPI Auto Setting Zero Position.....	24
7.6.8 SPI Single Byte Read Angle.....	24
7.6.9 SPI Burst Read Angle.....	26
8 Change & Program Zero Position to EEPROM.....	27
9 Calibration.....	28
9.1 Factory Calibration.....	28
9.2 User Auto-Calibration.....	29
9.3 User NLC Calibration.....	30
10 Register Map.....	31
11 Magnet Placement.....	38
12 Mechanical Angle Direction.....	39
13 Package Information.....	40
14 Copy Rights and Disclaimer.....	41
15 Revision History.....	42

1. Pin Configuration

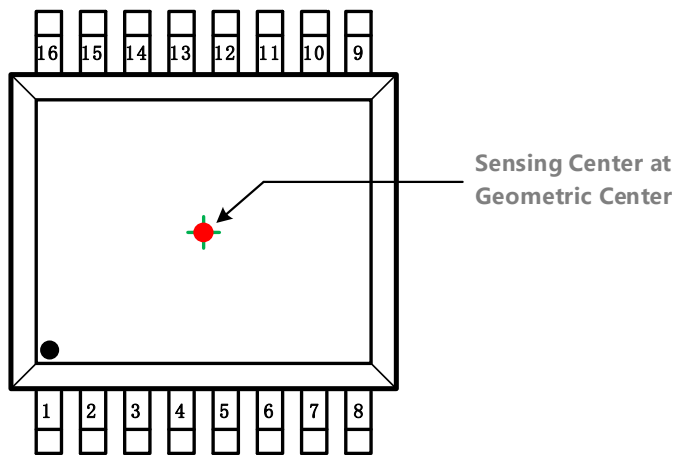


Figure 1: Pin Configuration of MT6835 (TSSOP-16) Package

I/O Pin List

I/O Name	#	I/O Type	Description
U	1	Digital Output	Commutation Signal U or –A
V	2	Digital Output	Commutation Signal V or –B
W	3	Digital Output	Commutation Signal W or -Z
CAL_EN	4	Digital Input	Auto-Calibration Enable Pin
MISO	5	Digital Output	SPI Data, Master Input Slave Output
MOSI	6	Digital Input	SPI Data, Master Output Slave Input
SCK	7	Digital Input	SPI Clock
CSN	8	Digital Input	SPI Chip Select
VDD	9	Power Supply	3.3~5.0V Power Supply
OUT	10	Digital Output	PWM Output
TEST	11	Analog Input	Factory Test Pin
VSS	12	Power Supply	Ground
TEST_EN	13	Digital Input	Factory Test Enable
Z	14	Digital Output	Incremental Signal Z
B	15	Digital Output	Incremental Signal B
A	16	Digital Output	Incremental Signal A

Family Members

Part Number	Description
MT6835GT-STD	TSSOP-16 Package, Tube (60 Pcs/Tube) or Reel (3000 Pcs/Reel)

*TSSOP-16 Reflow Sensitivity Classification: MSL-3

21-Bit High Accuracy Magnetic Angle Encoder IC

2. Functional Diagram

The MT6835 is manufactured in a CMOS standard process and uses advanced magnetic sensing technology to sense the magnetic field distribution across the surface of the chip. The integrated magnetic sensing element array is placed around the center of the device and delivers a voltage representation of the magnetic field at the surface of the IC.

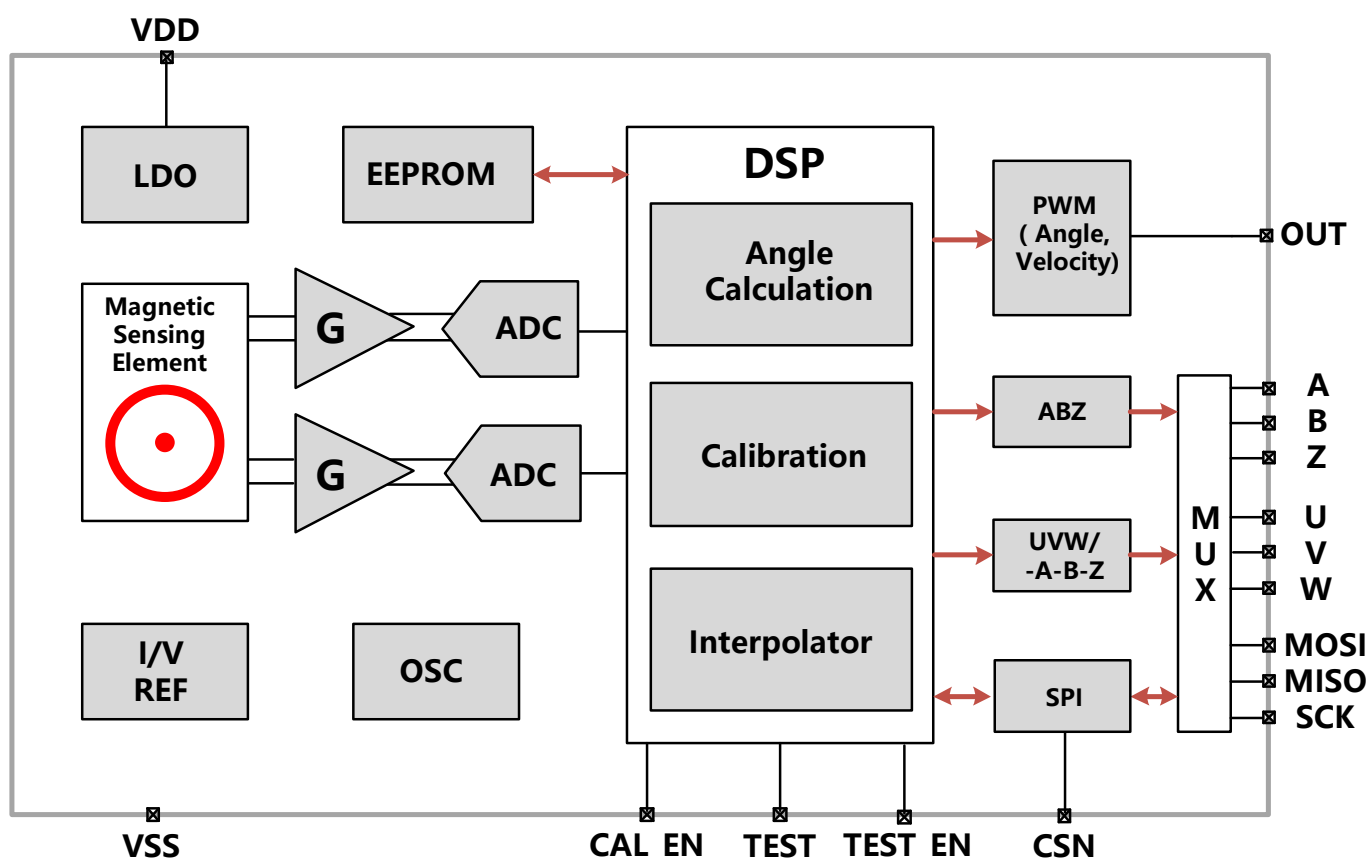


Figure 2: Block Diagram

Figure 2 shows a simplified block diagram of the chip, consisting of the magnetic sensing element modeled by two interleaved Wheatstone bridges to generate cosine and sine signals, gain stages, analog-to-digital converters (ADC) for signal conditioning, and a digital signal processing (DSP) unit for encoding. Other supporting blocks such as LDO, etc. are also included.

21-Bit High Accuracy Magnetic Angle Encoder IC

3. Absolute Maximum Ratings (Non-Operating)

Stresses beyond those listed under “Absolute Maximum Ratings” may cause permanent damage to the device. These are stress ratings only. Functional operation of the device at these or any other conditions beyond those indicated under “Operating Conditions” is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

Parameter	Min.	Max.	Unit	Notes
DC Voltage at Pin VDD	-0.5	6.5	V	
Terminal Voltage at Input and Output Pins	-0.5	VDD	V	ABZ, UVW, MISO, MOSI, SCK, CSN, TEST_EN, OUT, CAL_EN, Test
Output Current at Output Pins	-20	20	mA	ABZ, UVW, OUT, MISO
Storage Temperature	-40	150	°C	
I _{SCR} (Latch-up Input Current)	-	±100	mA	AEC-Q100-004
V _{HBM} (ESD Voltage)	-	±8.0	KV	AEC-Q100-002
V _{CDM} (ESD Voltage)	-	±1.5	KV	AEC-Q100-011

4. Operating Conditions

Parameter	Min.	Max.	Unit
DC Voltage at Pin VDD	3.0	5.5	V
Magnetic Flux Density Range	30	1,000	mT
Rotation Speed	-	120,000	RPM
Operating Temperature	-40	125	°C

21-Bit High Accuracy Magnetic Angle Encoder IC

5. Electrical Characteristics

Operation conditions: $T_a = -40$ to 125°C , $V_{DD} = 3.0 \sim 5.5\text{V}$ unless otherwise noted.

Symbol	Parameter	Conditions/Notes	Min.	Typ.	Max.	Unit
VDD	Supply Voltage	-	3.0	3.3~5.0	5.5	V
Idd	Supply Current	-	15	22	28	mA
LSB	Resolution (ABZ Mode)	N Steps per Cycle	-	$360^{\circ}/N$	-	$^{\circ}$
INL	Integral Non-Linearity with Factory Calibration	Note[1]	-	± 0.5	± 1.0	$^{\circ}$
	Integral Non-Linearity with User Auto-Calibration	Note[2]	-	± 0.07	-	$^{\circ}$
DNL	Differential Non-Linearity (ABZ Mode), Figure 3	@2500 PPR	-	± 0.005	-	$^{\circ}$
TN	Transition Noise (ABZ Mode)	25°C with 'BW' =5	-	0.0015	-	$^{\circ}\text{rms}$
Hyst	Hysteresis	Programmable	-	0.011	-	$^{\circ}$
T _{PwrUp}	Power-Up Time	VDD Ramp < 100us	-	64	-	ms
T _{Delay}	Propagation Delay	Constant Speed	-	10	-	us
T _{ST}	Step Response Time	Register 'BW' =5	-	100	-	us

Note[1]: The typical error value can be achieved at room temperature and with no off-axis misalignment error. The maximum error value can be achieved over operation temperature range, at maximum air gap and with worst-case off-axis misalignment error.

Note[2]: Please follow chapter 9 for the detail of 'User Auto-Calibration'.

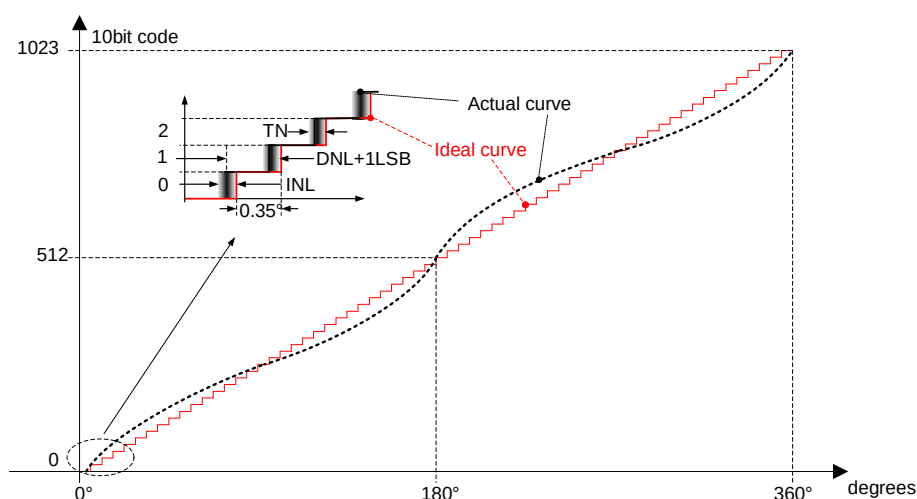


Figure 3: Drawing Illustration INL, DNL and TN (for 10-bit case)

21-Bit High Accuracy Magnetic Angle Encoder IC

ABZ Output Characteristics

Operation conditions: Ta=-40 to 125°C, VDD=3.0~5.5V unless otherwise noted.

Symbol	Parameter	Conditions/Notes	Min.	Typ.	Max.	Unit
AB _{RES}	AB Resolution	Programmable	1	-	16,384	Pulse/Round
AB _{Freq}	A or B Frequency	Note(3)	-	-	2.048	MHz

Note (3): The AB_{Freq} is the product of RS(Rotation Speed) and AB_{RES}. So, for maximum AB resolution of 16,384 PPR, the maximum supported rotation speed RS is:

$$RS_{MAX} = \frac{AB_{Freq}}{AB_{Res}} = \frac{2.048MHz}{16,384} = 125Hz = 7,500RPM$$

Even >2MHz AB_{Freq} is available, but the INL could not be guaranteed when AB_{Freq} is >2MHz

PWM Output Characteristics

Operation conditions: Ta=-40 to 125°C, VDD=3.0~5.5V unless otherwise noted.

Symbol	Parameter	Conditions/Notes	Min.	Typ.	Max.	Unit
F _{PWM}	PWM Frequency	Programmable	-8%	497/994	+8%	Hz
T _{Rise}	Rising Time	C _L =1nF	-	-	1	us
T _{Fall}	Falling Time	C _L =1nF	-	-	1	us

EEPROM Characteristics

Operation conditions: Ta=-40 to 125°C, VDD=3.0~5.5V unless otherwise noted.

Symbol	Parameter	Min.	Typ.	Max.	Unit
VDD	Supply Voltage @Erase/Program EEPROM	3.0	-	5.5	V
Endurance	Erase/Program Cycles	1,000	-	-	Cycles
Retention	Data Retention @150°C	10	-	--	Years

21-Bit High Accuracy Magnetic Angle Encoder IC

Digital I/O Characteristics

Operation conditions: Ta=-40 to 125°C, VDD=3.0~5.5V unless otherwise noted.

Symbol	Parameter	Conditions/Notes	Min.	Typ.	Max.	Unit
V _{IH}	Digital I/O Input Voltage High Level Voltage (CSN, MOSI, CLK, CAL_EN)	-	0.8*VDD	-	-	V
V _{IL}	Digital I/O Input Voltage Low Level Voltage (CSN, MOSI, CLK, CAL_EN)	-	-	-	0.2*VDD	V
V _{OH}	Digital I/O Output High Level Voltage (ABZ, UVW, MISO, OUT)	Push-Pull (@Iout=-2mA)	VDD-0.4	-	-	V
V _{OL}	Digital I/O Output Low Level Voltage (ABZ, UVW, MISO, OUT)	Push-Pull (@Iout=2mA)	-	-	0.4	V
T _{Rise}	Rising Time	C _L =20pf	-	-	50	ns
T _{Fall}	Falling Time	C _L =20pf	-	-	50	ns

21-Bit High Accuracy Magnetic Angle Encoder IC

6. Magnetic Input Specifications

Operation conditions: Ta=-40 to 125°C, VDD=3.0~5.5V unless otherwise noted, two-pole cylindrical diametrically magnetized source.

Symbol	Parameter	Conditions/Notes	Min.	Typ.	Max.	Unit
Dmag	Diameter of Magnet	Recommended Magnet: Ø10mm x 2.5mm for Cylindrical Magnets	-	10	-	mm
Tmag	Thickness of Magnet	-	-	2.5	-	mm
Bpk	Magnetic Input Field Amplitude	Measure at the IC Surface	30	-	1,000	mT
AG	Air Gap	Magnetic to IC Surface Distance	-	1.0	3.0	mm
RS	Rotation Speed	Turn Rounds per Minute	-	-	120,000	RPM
DISP	Off Axis Misalignment	Misalignment Error Between Sensor Sensing Center and Magnet Axis (See Figure 4)	-	-	0.3	mm
TCmag1	Recommended Magnet Material and Temperature Drift Coefficient	NdFeB (Neodymium Iron Boron)	-	-0.12	-	% / °C
TCmag2		SmCo (Samarium Cobalt)	-	-0.035	-	

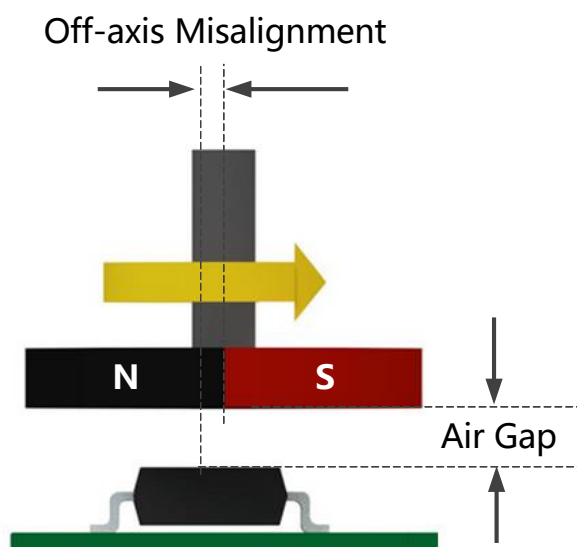


Figure 4: Magnet Arrangement

21-Bit High Accuracy Magnetic Angle Encoder IC

7. Output Mode

The MT6835 provides ABZ, UVW and PWM signals at output pins, and the 21-bit absolute angle position data could be transferred by the standard 4-Wire SPI interface.

7.1 I/O Pin Configuration

For TSSOP-16 package, ABZ (Single-end or differential), UVW, PWM and SPI Interface are configured as below table.

I/O Pin Configuration

Pin#	UVW+SPI+PWM+ABZ	-A-B-Z+SPI+PWM+ABZ
1	U	-A
2	V	-B
3	W	-Z
5	MISO	MISO
6	MOSI	MOSI
7	SCK	SCK
8	CSN	CSN
10	OUT(PWM)	OUT(PWM)
14	Z	Z
15	B	B
16	A	A

21-Bit High Accuracy Magnetic Angle Encoder IC

7.2 Reference Circuit

The MT6835 is powered by a single supply VDD (3.3~5.0V), so a decoupling capacitor no less than 0.1uf between VDD and GND is necessary. For better EMC performance, we highly recommend adding a TVS between VDD and GND. The default reference circuit is shown in Figure. 5.

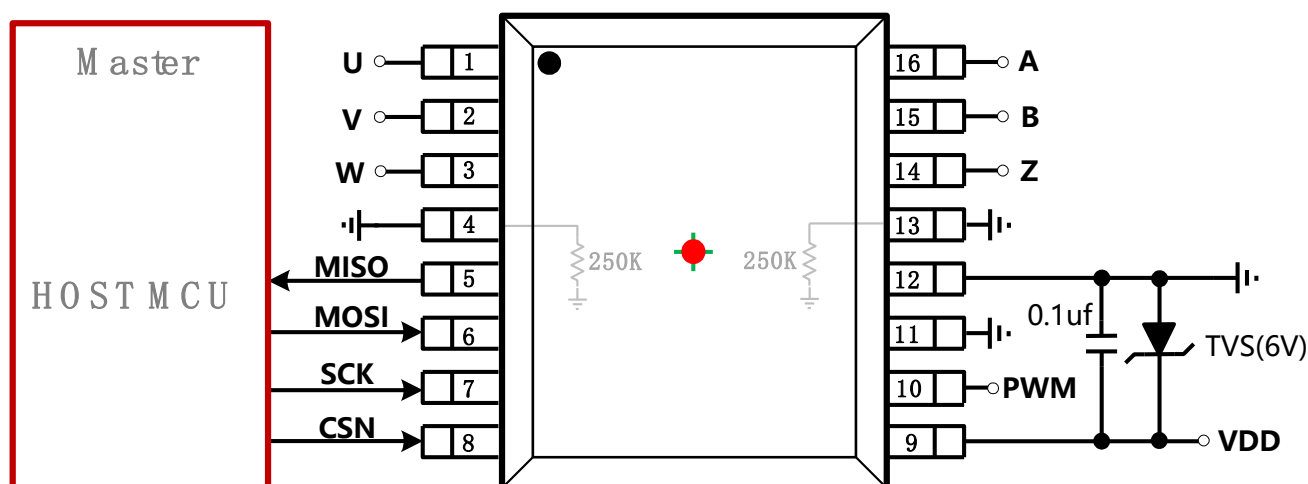


Figure 5: Reference Circuit without User Auto-Calibration

The reference circuit for User Auto-Calibration is shown in Figure. 6, the detail of User Auto-Calibration please refer to Chapter 9.2.

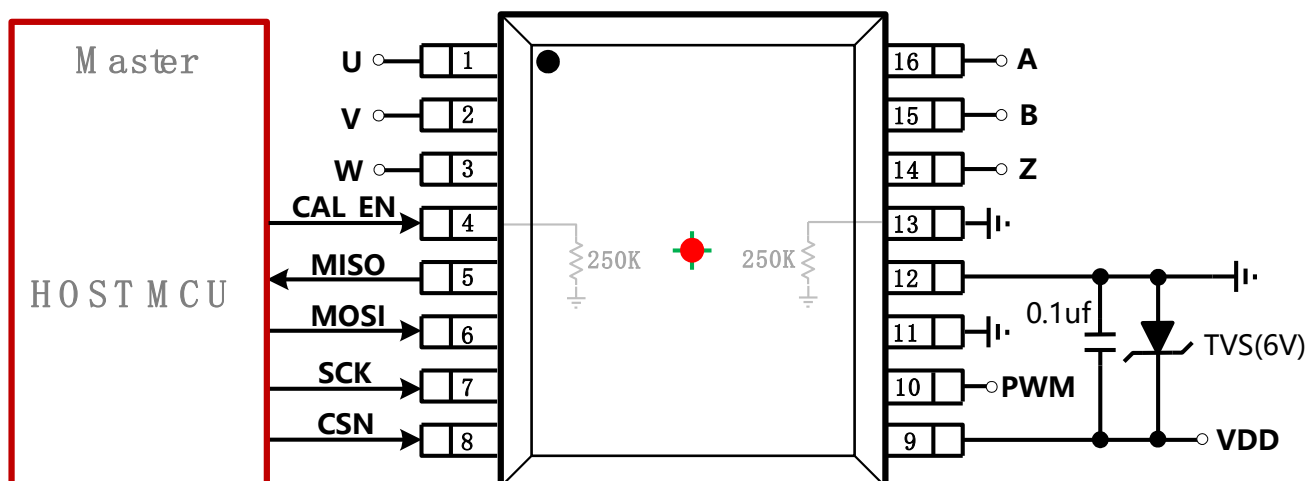


Figure 6: Reference Circuit with User Auto-Calibration

7.3 Quadrature A,B and Zero-Position Output (ABZ Mode)

As shown in Figure 7, when the magnet rotates counter-clock-wise (CCW), the rising edge of output B leads output A by 1/4 cycle, when the magnet rotates clock-wise (CW), the rising edge of output A leads output B by 1/4 cycle (or 1 LSB). Output Z indicates the zero position of the magnet. After chip power-on, the ABZ output is blocked for 64ms to guarantee proper output.

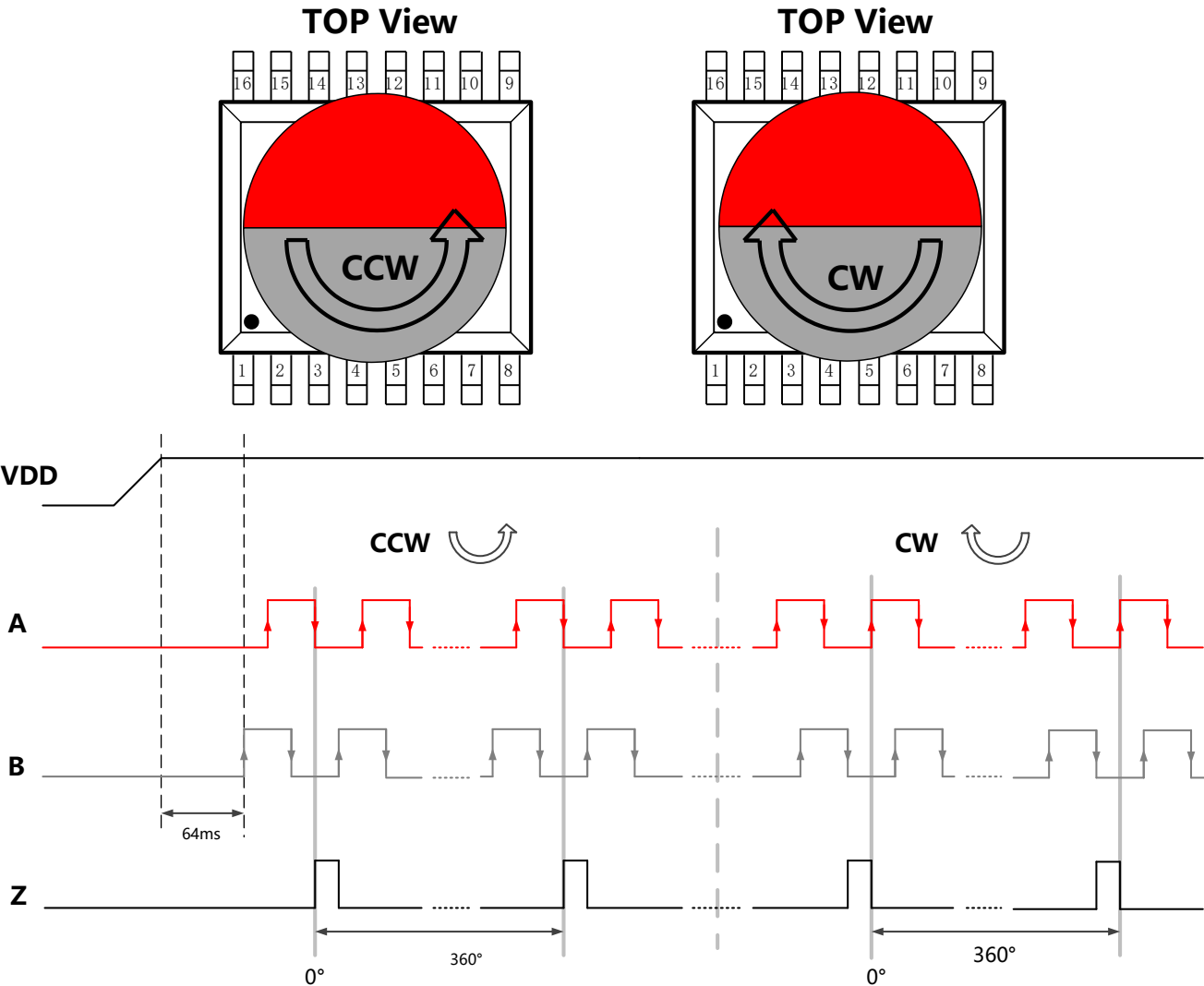


Figure 7: ABZ output with VDD power on

‘ROT_DIR’ (CCW or CW) Register (EEPROM)

Reg. ROT_DIR	Rotation Direction
0x0	Counter-Clock-wise, Output B leads Output A
0x1	Clock-wise, Output B leads Output A

‘ ROT_DIR ’ is effective for all output types as ABZ, UVW, PWM and SPI Angle data

21-Bit High Accuracy Magnetic Angle Encoder IC

ABZ resolution is user programmable from 1~16,384 PPR. The relationship between binary bits, LSBs and PPR resolution of ABZ output are shown in Figure 8 & Figure 9.

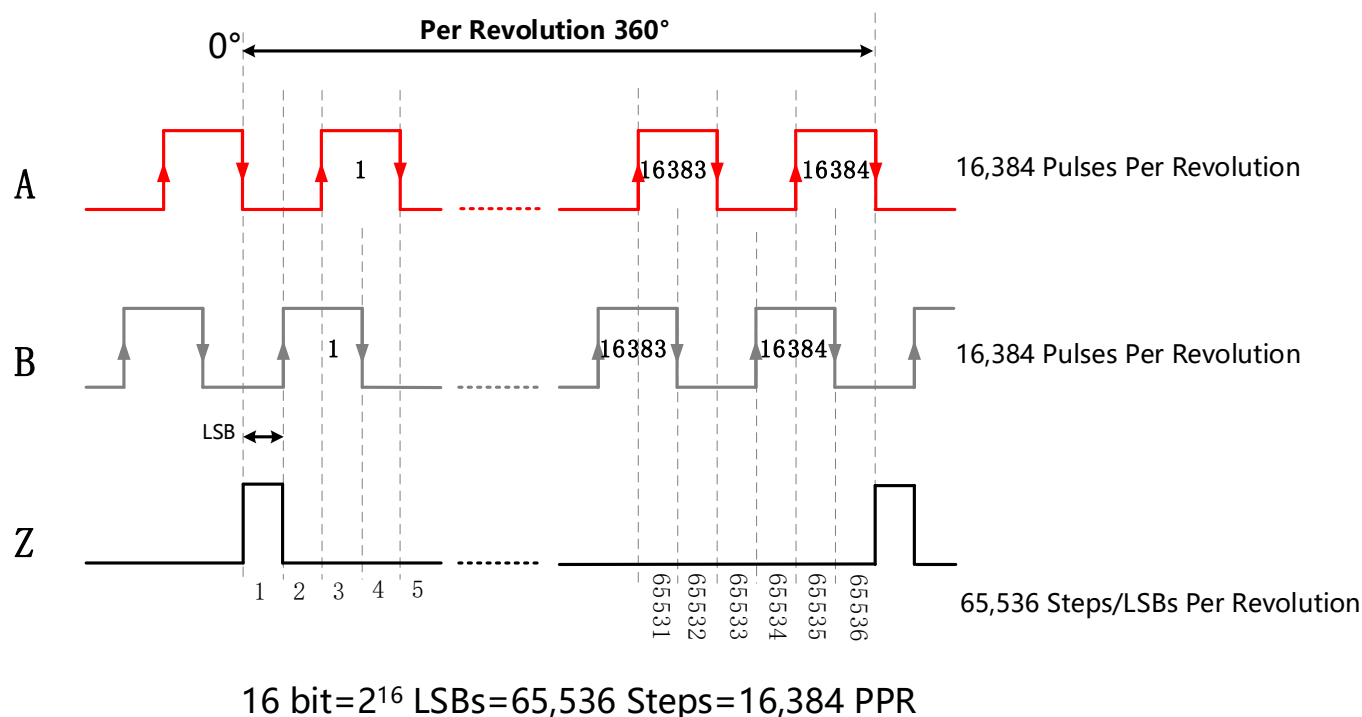


Figure 8: ABZ Output Resolution = 16 bit = 16,384 PPR

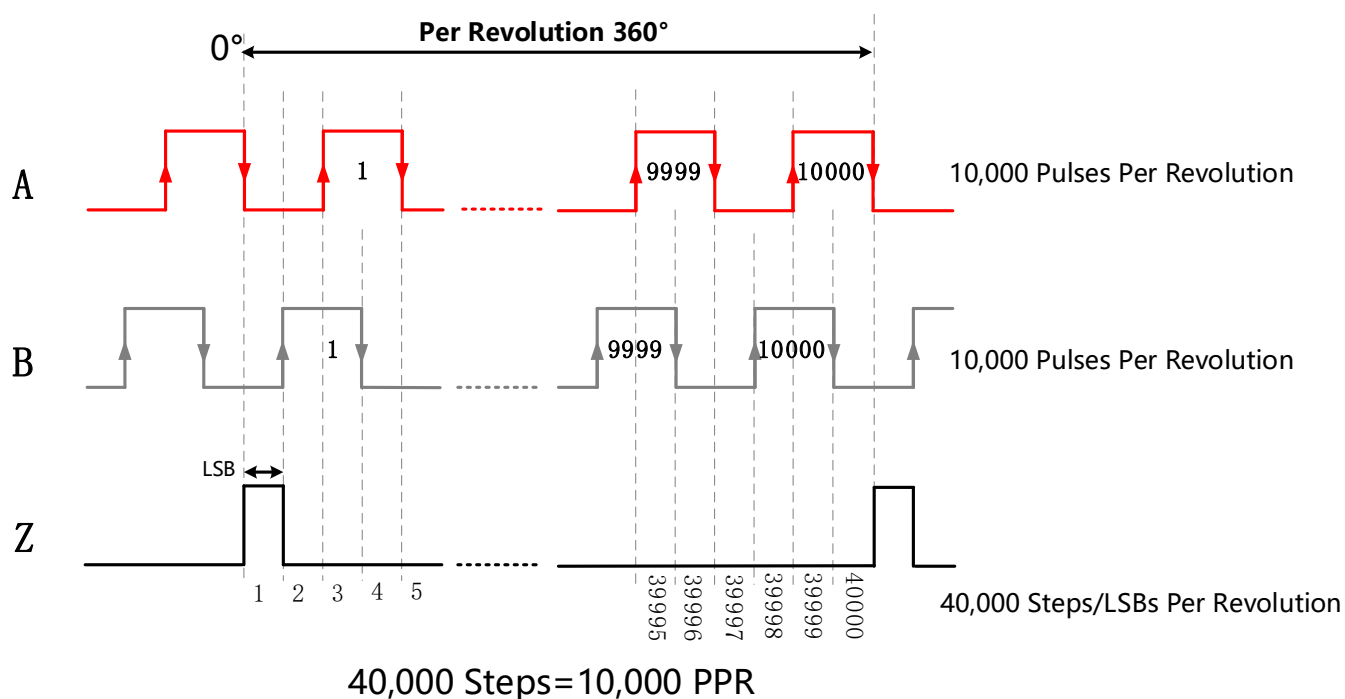


Figure 9: ABZ Output Resolution = 10,000 PPR

21-Bit High Accuracy Magnetic Angle Encoder IC

The resolution of ABZ is defined by a 14-bit register 'ABZ_RES[13:0]' ;

'ABZ_RES[13:0]' Register (EEPROM)

Reg. ABZ_RES[13:0]	AB Resolution (Pulse per. Round)
0x0000	1
0x0001	2
0x0002	3
⋮	⋮
0x3FFC	16,381
0x3FFD	16,382
0x3FFE	16,383
0x3FFF	16,384

Output Z indicates the zero position of the magnet which is user programmable, and the pulse width of Z is selectable as 1, 2, 4, 8, 16 LSBs or 60°, 120°, 180° as shown in Figure 10 and Figure 11. It is guaranteed that one Z pulse is generated for every rotation.

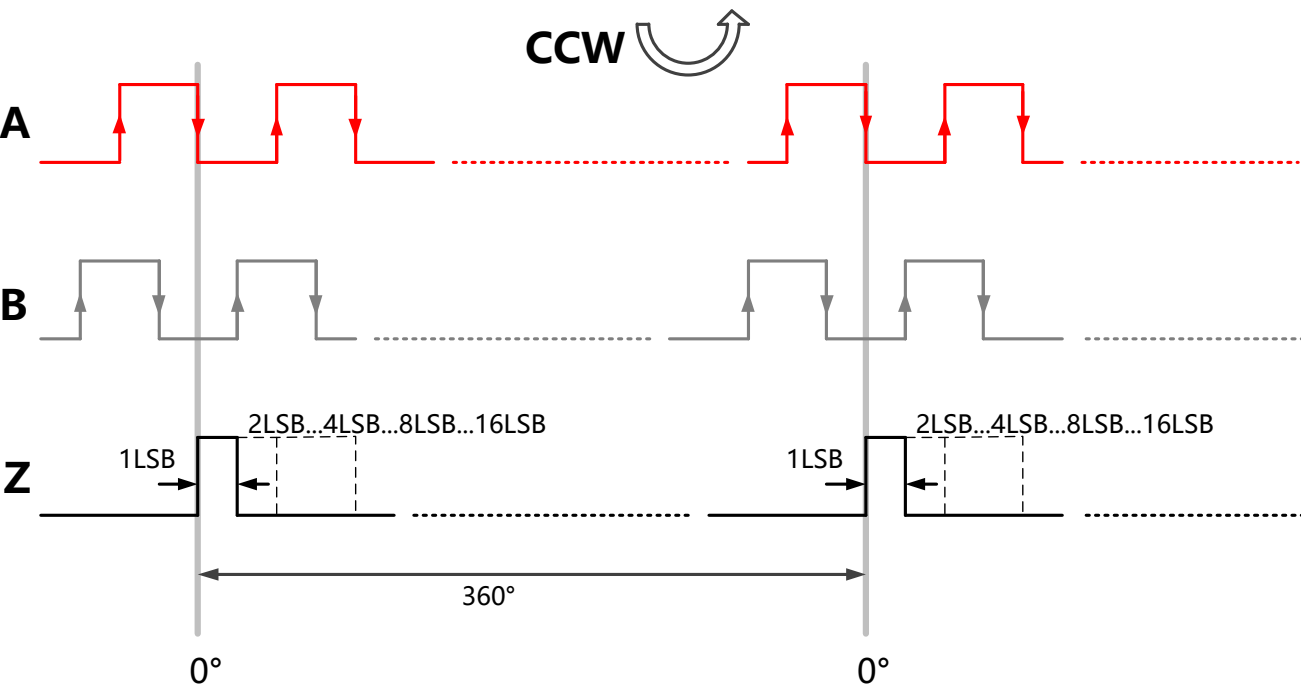


Figure 10: Typical ABZ Output with Z pules width=1,2,4,8 and 16 LSBs

21-Bit High Accuracy Magnetic Angle Encoder IC

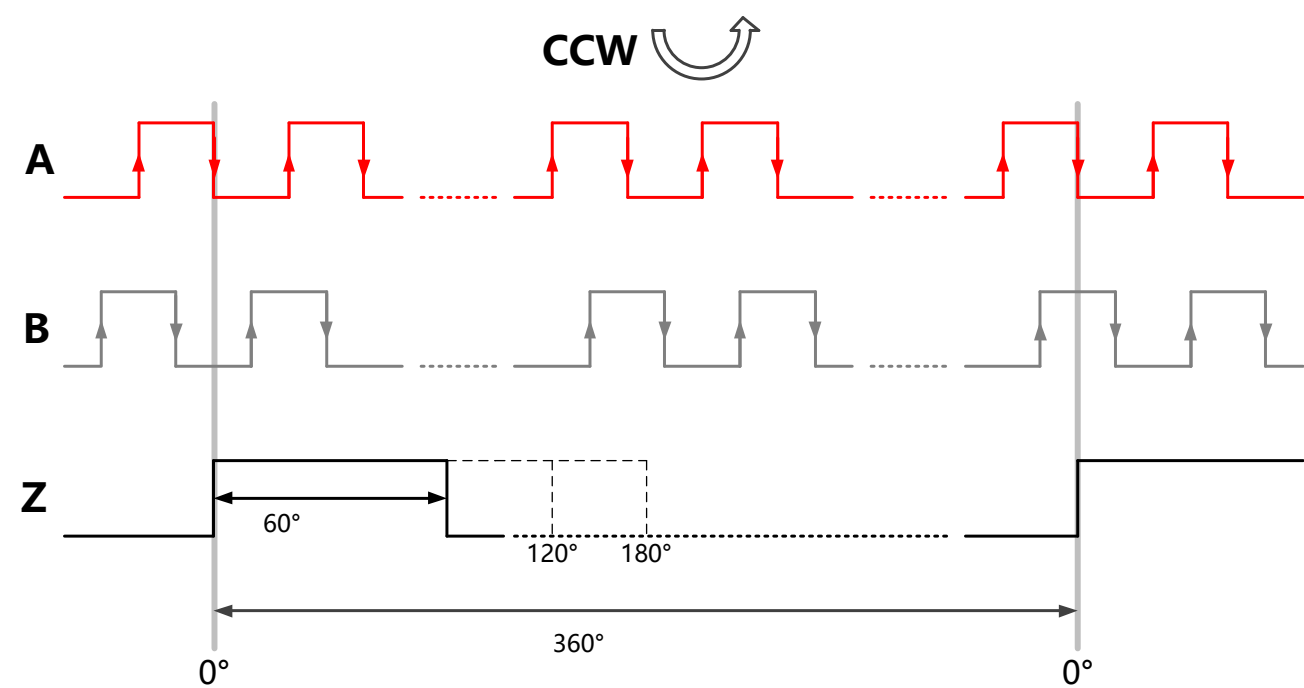


Figure 11: Typical ABZ Output with Z pules width=60°, 120° and 180°

The width of Z pulse is defined by the 3-bit register 'Z_PUL_WID[2:0]' ;

'Z_PUL_WID[2:0]' Register (EEPROM)

Reg. Z_PUL_WID[2:0]	Width (LSBs/°)	Reg. Z_PUL_WID[2:0]	Width (LSBs/°)
0x0	1	0x4	16
0x1	2	0x5	60°
0x2	4	0x6	120°
0x3	8	0x7	180°

The absolute position of Z pulse is defined by the 12-bit register 'ZERO_POS[11:0]' ;

'ZERO_POS[11:0]' Register (EEPROM)

Reg. ZERO_POS[11:0]	Absolute Position (°)
0x000	0
0x001	0.088
0x002	0.176
⋮	⋮
⋮	⋮
⋮	⋮
0xFFE	359.824
0xFFF	359.912

21-Bit High Accuracy Magnetic Angle Encoder IC

Also, Z pulse phase could be user programmable by 'Z_PHASE[1:0]' register as shown in Figure 12.

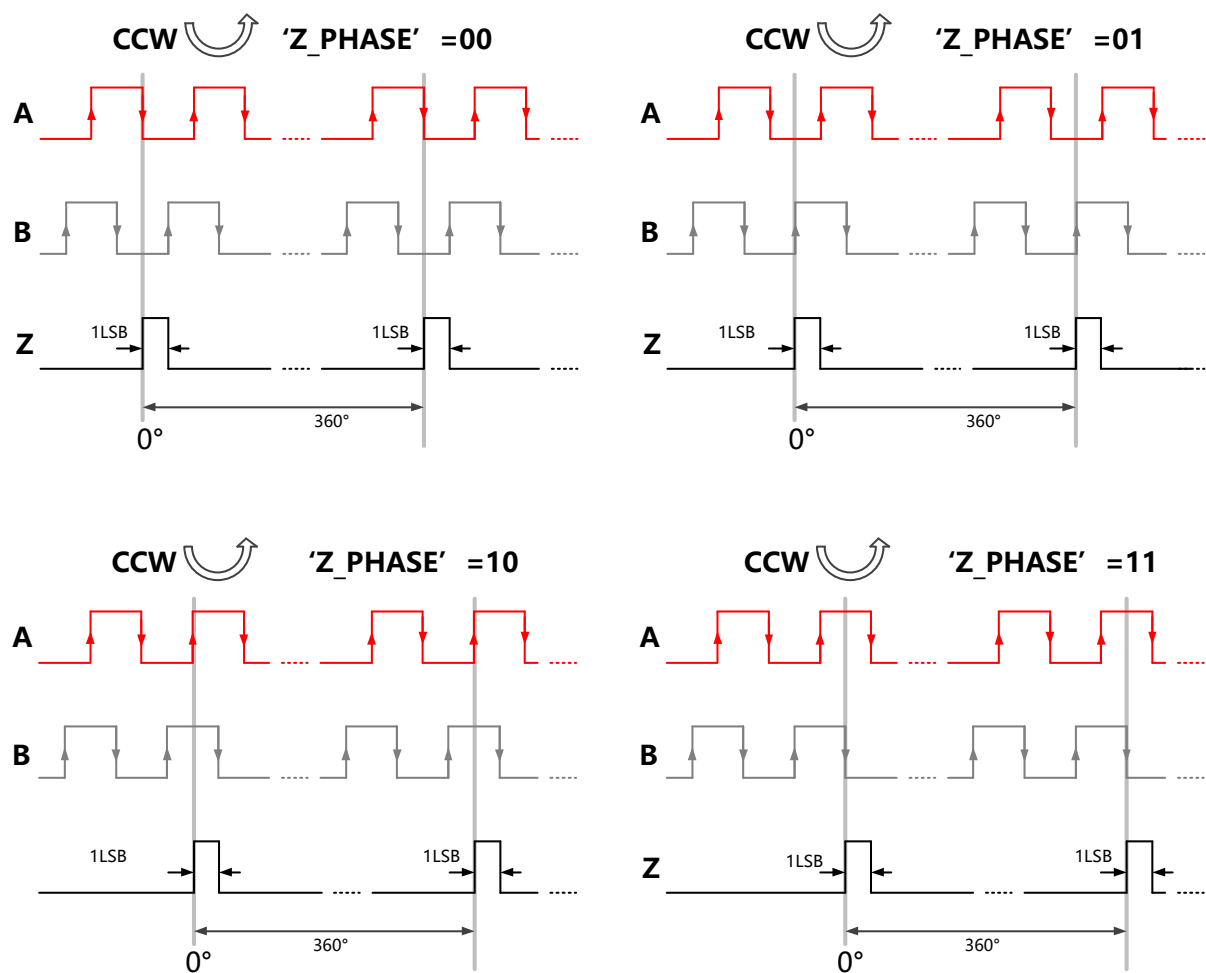


Figure 12: Z Pulse Phase with 'ROT_DIR' = 0, 'SWAP_AB' = 0, 'Z_EDGE' = 0

The relationship of 0° and Z pulse edge is defined by register 'Z_EDGE'

'Z_EDGE' Register (EEPROM)

Reg. Z_EDGE	Description
0x0	Z Pulse Rising Edge Aligned with Zero-Degree
0x1	Z Pulse Falling Edge Aligned with Zero-Degree

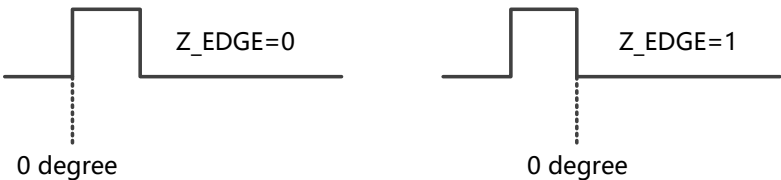


Figure 13: Z Pulse Edge with 0 degree

7.4 UVW Output Mode

The MT6835 provides U, V and W pulses which are 120° (electrical angle) out of phase as shown in Figure 14. The cycles of UVW per rotation can be programmed.

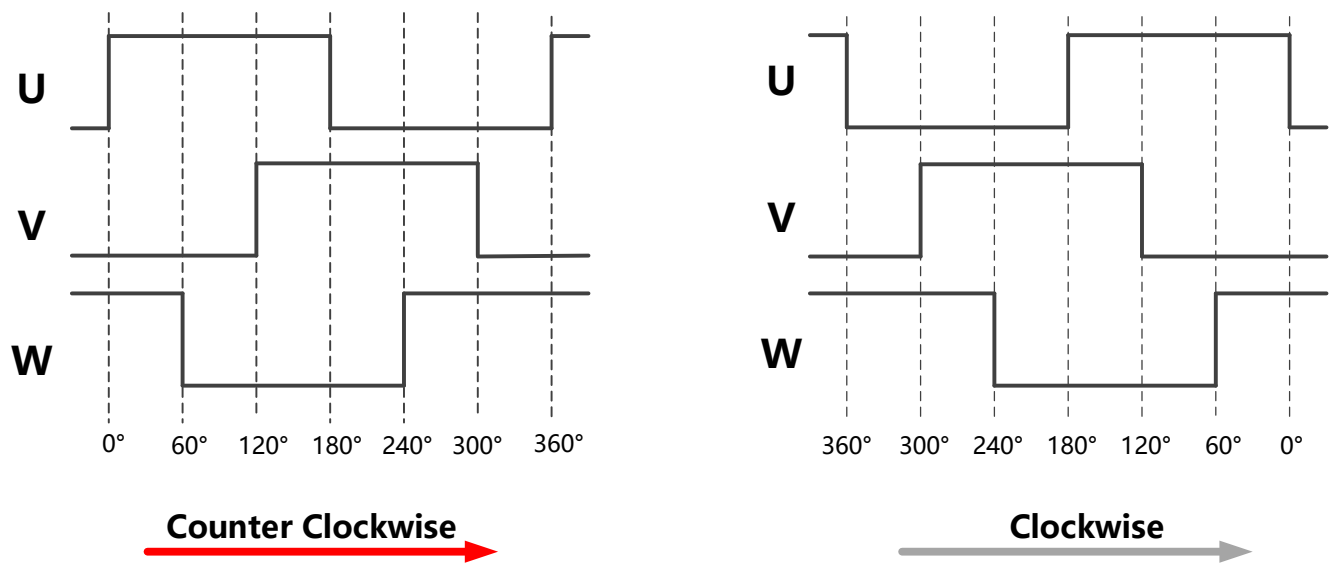


Figure 14: Typical Output Waveform for UVW Mode

‘UVW_RES’ Register (EEPROM)

Reg. UVW_RES[3:0]	UVW Pole Pairs
0x0	1
0x1	2
0x2	3
0x3	4
0x4	5
0x5	6
0x6	7
0x7	8
0x8	9
0x9	10
0xA	11
0xB	12
0xC	13
0xD	14
0xE	15
0xF	16

21-Bit High Accuracy Magnetic Angle Encoder IC

7.5 Pulse Width Modulation (PWM) Output Mode

The MT6835 provides a digital Pulse Width Modulation (PWM) output, whose duty cycle is proportional to the measured angle as shown in Figure 15. PWM is a default output of Pin.10.

The PWM output consists of a frame of 4119 PWM clock periods. The angle data is represented with 12-bit resolution in the frame. One PWM clock period represents 0.088° and has a typical duration of 244ns which also could be programmed to be 488ns.

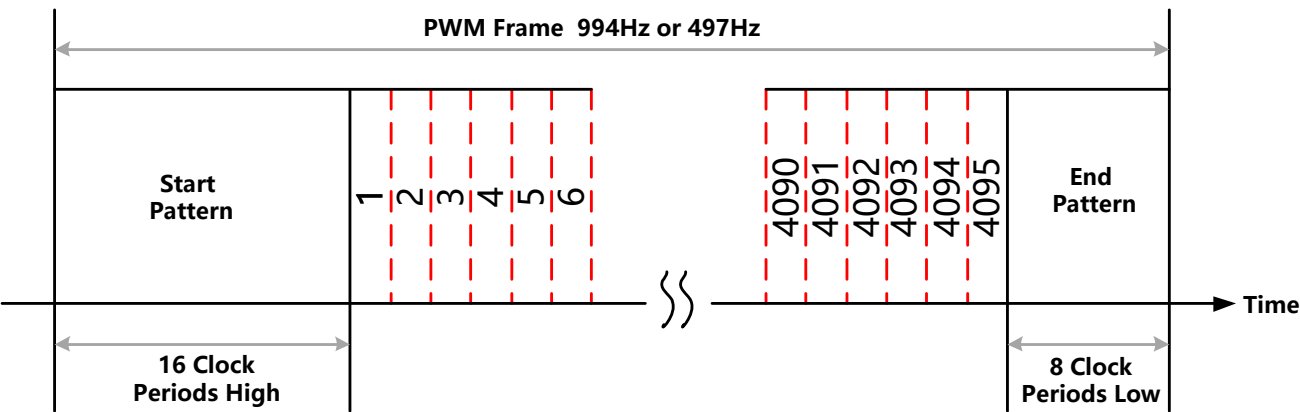


Figure 15: PWM Output Frame

‘PWM_FQ’ Register (EEPROM)

Reg. PWM_FQ	PWM Frame Frequency
0x0	994 Hz
0x1	497 Hz

‘PWM_SEL[2:0]’ Register (EEPROM)

Reg. PWM_SEL[2:0]	PWM Data Source
0x0	12-bit Angle Data
0x2	12-bit Velocity Data
Others	Factory Test Data

21-Bit High Accuracy Magnetic Angle Encoder IC

7.6 SPI Interface

The MT6835 also provides a 4-Wire SPI interface for a host MCU both to read back digital absolute angle information from its internal registers and to program its EEPROM.

7.6.1 SPI Reference Circuit

The reference circuit for SPI interface of a single chip please refer to Figure 5 and Figure 6. The multi-chip application is shown in Figure 16.

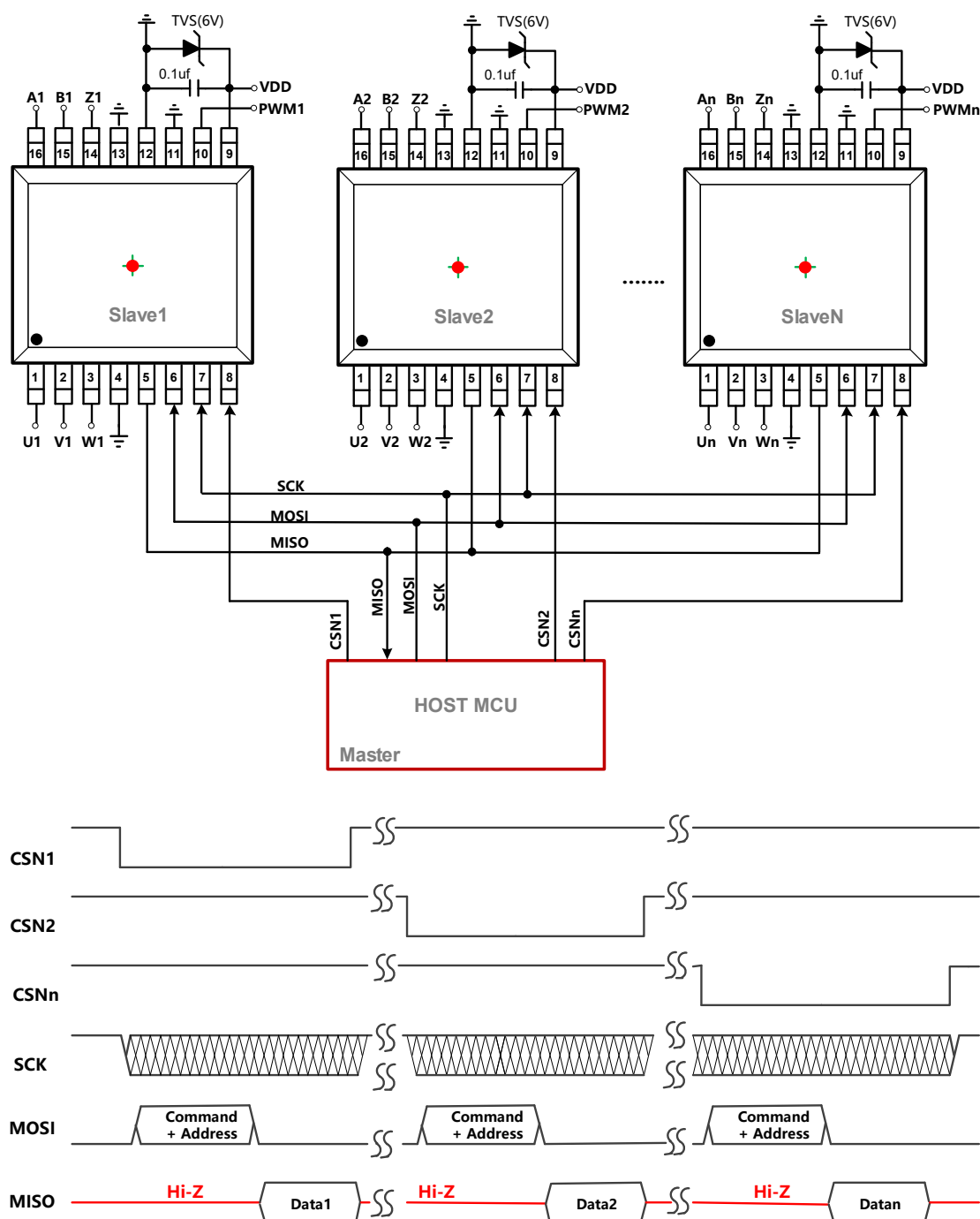


Figure 16: 4-Wire SPI Reference Circuit with multi-chips

21-Bit High Accuracy Magnetic Angle Encoder IC

7.6.2 SPI Timing Diagram

The MT6835 SPI uses mode3 (CPOL=1, CPHA=1) to exchange data. As shown in Figure 17, a data transfer starts with the falling edge of CSN. The MT6835 samples data on the rising edge of SCK, and the data transfer finally stops with the rising edge of CSN.

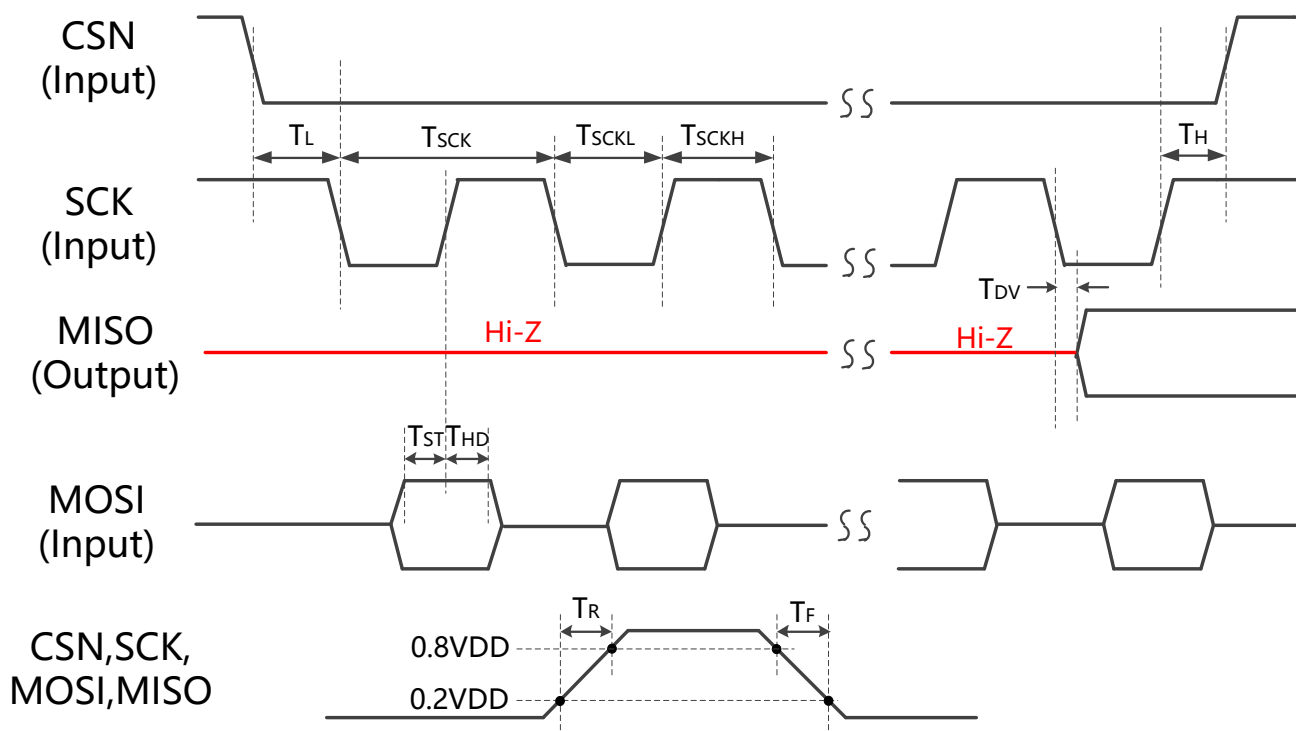


Figure 17: SPI Timing Diagram

SPI Timing Parameter

Symbol	Notes	Min.	Typ.	Max.	Unit
T_L	Time Between CSN Falling Edge and SCK Falling Edge	100		-	ns
T_{SCK}	Clock Period	60		-	ns
T_{SCKL}	Low Period of Clock	30		-	ns
T_{SCKH}	High Period of Clock	30		-	ns
T_H	Time Between SCK Last Rising Edge and CSN Rising Edge	$0.5 \cdot T_{SCK}$		-	ns
T_R	Rise Time of Digital Signal (with 20pf Loading Condition)	-	10	-	ns
T_F	Fall Time of Digital Signal (with 20pf Loading Condition)	-	10	-	ns
T_{DV}	Data Valid Time of MISO (with 20pf Loading Condition)	-	-	15	ns
T_{ST}	Setup Time of MOSI Data	10	-	-	ns
T_{HD}	Hold Time of MOSI Data	10	-	-	ns

21-Bit High Accuracy Magnetic Angle Encoder IC

7.6.3 SPI Protocol

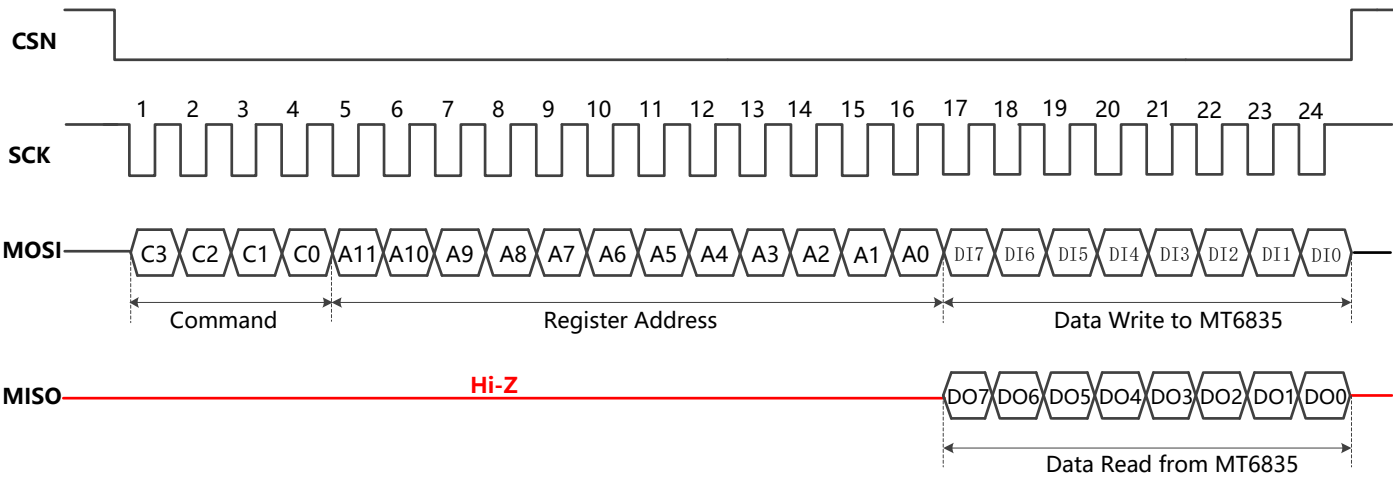


Figure 18: 4-Wire SPI Timing

An SPI data transfer starts with the falling edge of CSN and stops at the rising edge of CSN. SCK is the Serial Port Clock, and it is controlled by the SPI master, it is high when there is no SPI transmission. MOSI (master output slave input) and MISO (master input slave output) is the Serial Port Data Input and Output, it is driven at the falling edge of SCK and should be captured at the rising edge of SCK. The ‘MISO’ keeps Hi-Z unless it drives data as shown in Figure 18.

Bit 1-4: Operation Command bits C3~C0.

C3~C0	Operation	Notes
0011	Read	User Read Registers
0110	Write	User Write Registers
1100	Program EEPROM	User Erase and Program EEPROM
0101	Auto Setting Zero-Position	Auto Setting Current Position as Zero Position to Register
1010	Burst Angle Read	Read Angle Registers Continuously
Others	N/A	-

Bit 5-16: Address A11~A0. This is the address field of the indexed register.

Bit 17-24: Data DI7~DI0 (Write Operation). This is the data that will be written into the device.

Bit 17-24: Data DO7~DO0 (Read Operation). This is the data that will be read from the device.

21-Bit High Accuracy Magnetic Angle Encoder IC

7.6.4 SPI Read One Byte Register

For single byte read, the operation command C3~C0= '0011' , and the target 12-bit register address A11~A0 please refer to Chapter 10.

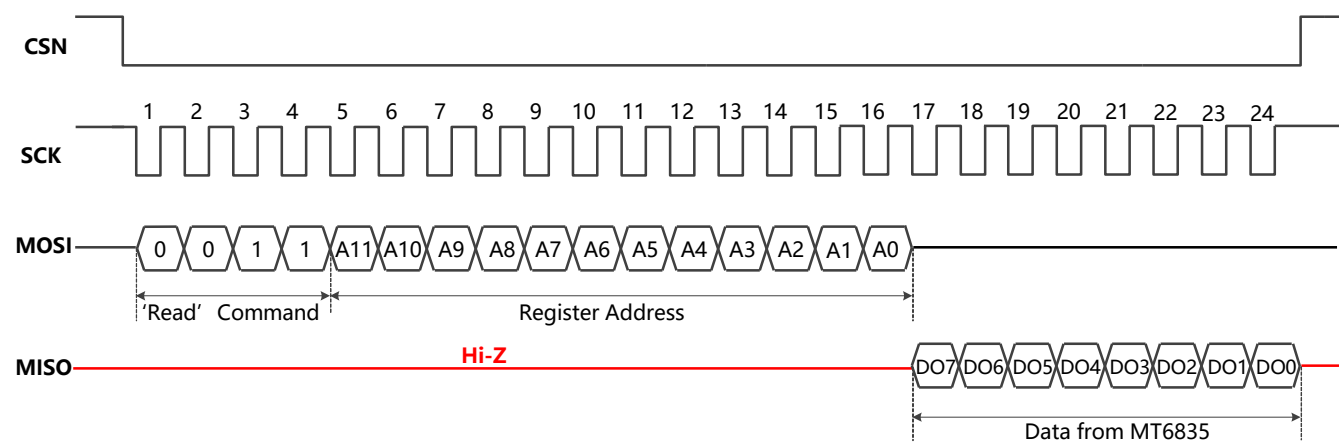


Figure 19: SPI Read One Byte Register

7.6.5 SPI Write One Byte Register

For single byte write, the operation command C3~C0= '0110' , and the target 12-bit register address A11~A0 please refer to Chapter 10.

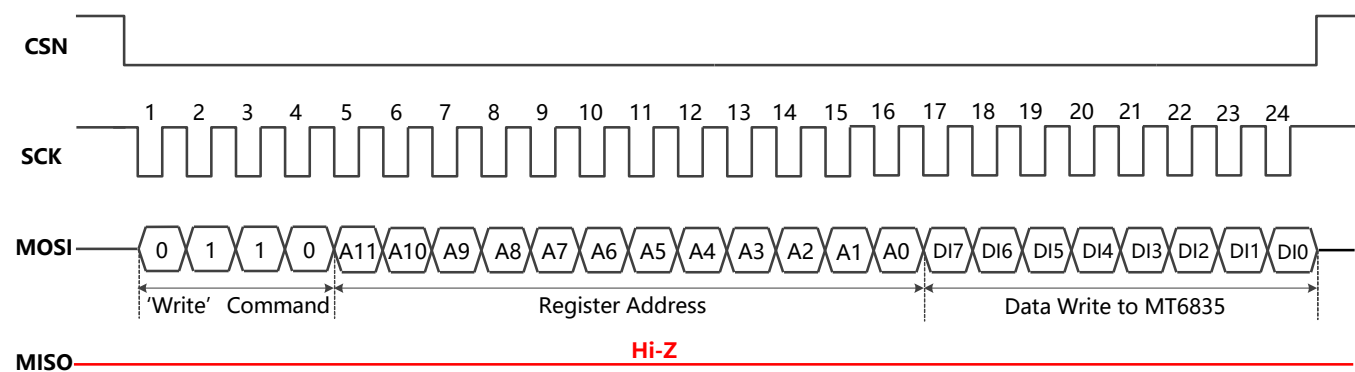


Figure 20: SPI Write One Byte Register

21-Bit High Accuracy Magnetic Angle Encoder IC

7.6.6 SPI Program EEPROM

For EEPROM programming, the operation command C3~C0= '1100' , and all the data in 'Register Map' will be programmed to EEPROM. When the command received successfully, the acknowledge bits ack7~ack0 will return value 0x55; any other value indicates the command received failed. Please **wait at least 6 seconds** before power-down MT6835 after EEPROM programming.

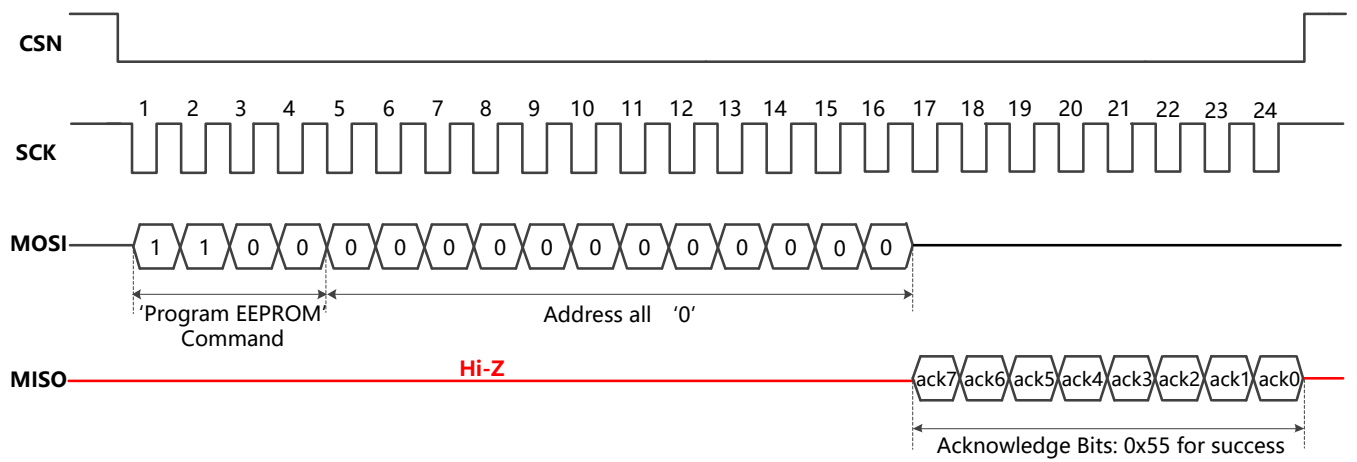


Figure 21: SPI Program EEPROM

When the data is programmed to EEPROM, they will be non-volatile; while the data in register map it is volatile, meaning it is lost when the power goes off. The user could read/write the register map through SPI interface and program the data of register map into EEPROM by SPI command.

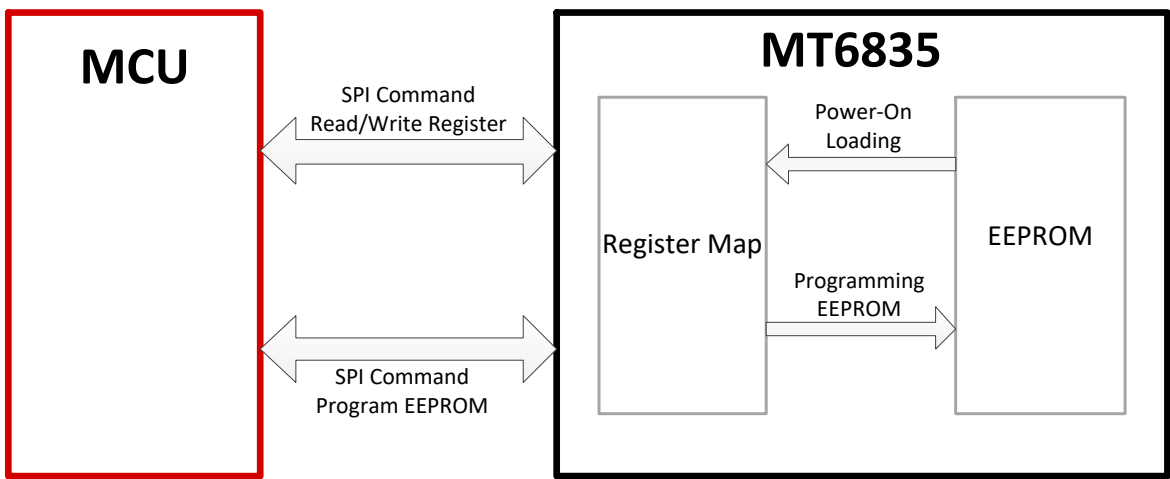


Figure 22: Register Map and EEPROM

21-Bit High Accuracy Magnetic Angle Encoder IC

7.6.7 SPI Auto Setting Zero Position

MT6835 provide an Auto Setting Zero Position command, which will automatically write current position as the new Zero Position to the register 'ZERO_POS[11:0]'. The operation command is C3~C0= '0101', when the command is received successfully, the acknowledge bits ack7~ack0 will return value 0x55; any other value indicates the command received failed.

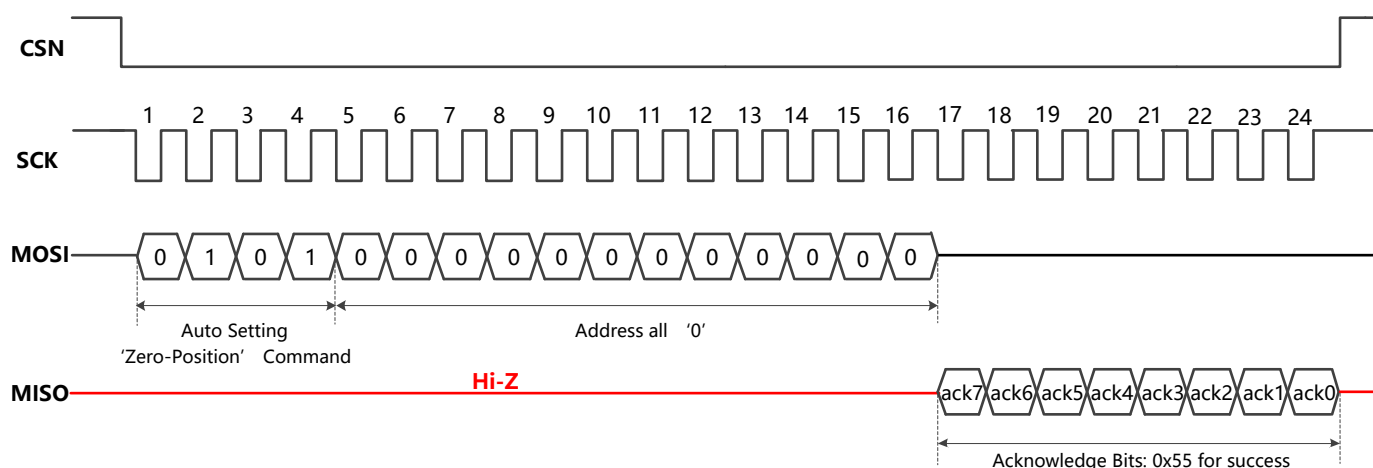


Figure 23: SPI Auto Setting Zero-Position Register

This command only stores the current position into 'ZERO_POS[11:0]' register. If the user want to program it to EEPROM, an SPI program EEPROM operation (waiting >1ms after 'Auto Setting Zero Position' command) should be additionally done as shown in Figure 21.

7.6.8 SPI Single Byte Read Angle Registers

The 21-bit absolute angle data could be read by SPI interface as shown in Figure 24. In order to facilitate the user to synchronize the sampling of angle data, when CSN is pulled down, MT6835 internally latched the data of 0x003~0x006, which will not be refreshed until all 0x003~0x006 registers have been read out.

21-Bit High Accuracy Magnetic Angle Encoder IC

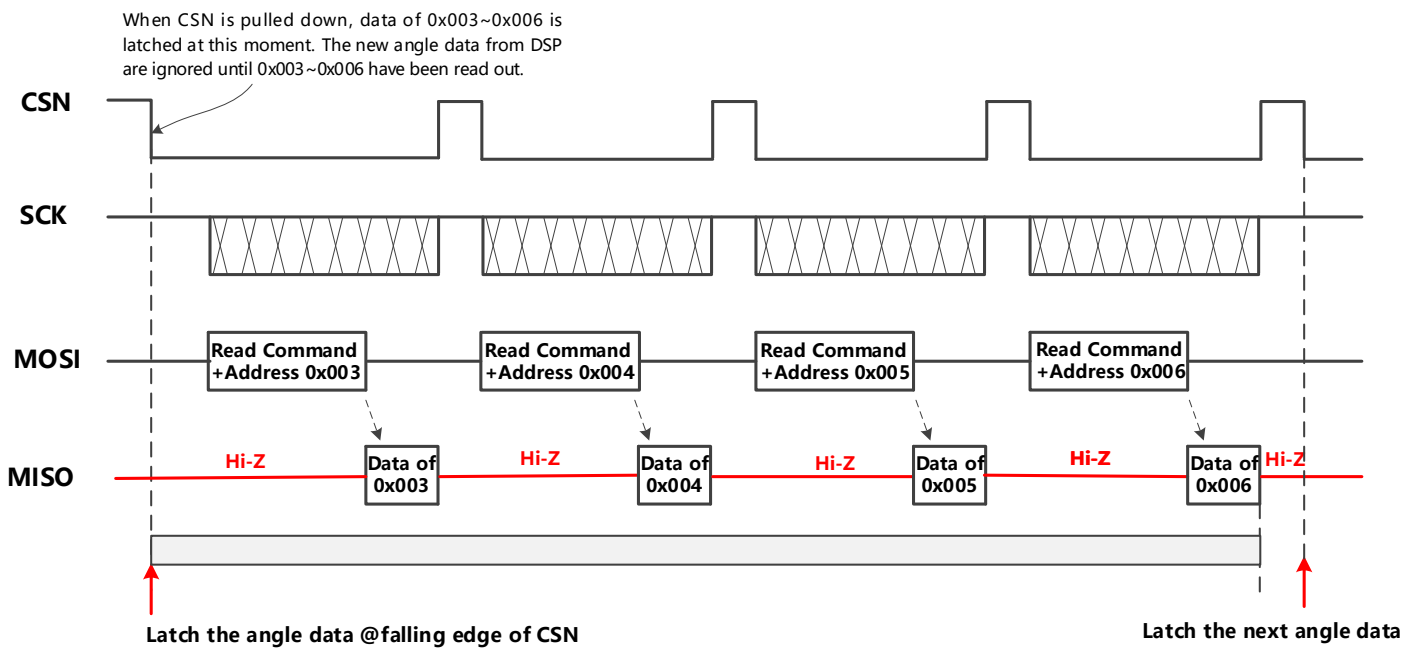


Figure 24: SPI Single Byte Read Angle Register

‘ANGLE[20:0]’ Angle Data Register (Read Only)

Address	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0x003	ANGLE[20:13]							
0x004	ANGLE[12:5]							
0x005	ANGLE[4:0]					STATUS[2:0]		
0x006	CRC[7:0]							

0~360° absolute angle θ could be calculated by the below formula with ANGLE[20:0]:

$$\theta = \frac{\sum_{i=0}^{20} \text{ANGLE} \langle i \rangle \cdot 2^i}{2^{21}} \cdot 360^\circ$$

‘STATUS[2:0]’ Register (Read Only)

STATUS[2:0]	Notes
Bit [0]	Rotation Over Speed Warning; Default ‘0’ , Logic ‘1’ for Warning
Bit [1]	Weak Magnetic Field Warning; Default ‘0’ , Logic ‘1’ for Warning
Bit [2]	Under Voltage Warning; Default ‘0’ , Logic ‘1’ for Warning

CRC Data Range: ANGLE[20:0] and STATUS[2:0] total 24bits

CRC polynomial: X^8+X^2+X+1 , MSB (ANGLE[20]) shifts in first.

21-Bit High Accuracy Magnetic Angle Encoder IC

7.6.9 SPI Burst Read Angle Registers

The MT6835 provides an SPI burst read angle registers mode for faster data transfer than single byte read mode as shown in Figure 25. The operation command of this mode is C3~C0= '1010' , after MCU sends this command with address 0x003, MT6835 continuously outputs angle data of register 0x003~0x006.

In order to facilitate the user to synchronize the sampling of angle data, when CSN is pulled down, MT6835 internally latches the data of 0x003~0x006, which will not be refreshed until all 0x003~0x006 registers have been read out.

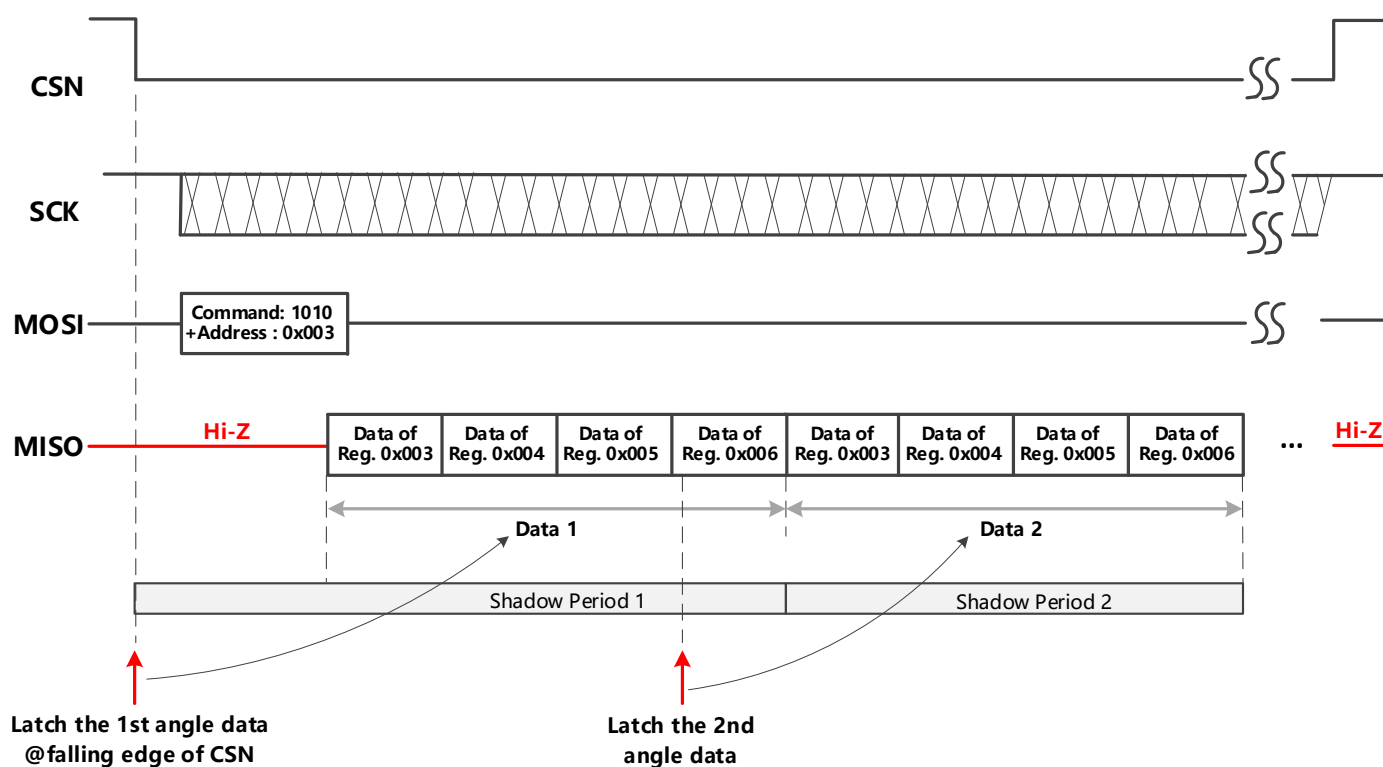


Figure 25: SPI Burst Read Angle Registers

8. Change & Program Zero-Position to EEPROM

The 'ZERO_POS[11:0]' register defines the zero degree of MT6835, the default zero degree of MT6835 with a two-pole magnet is shown in Chapter 12. There are two methods to change the register 'ZERO_POS[11:0]' as shown in Figure 26.

Address	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0	Type
0x009	ZERO_POS[11:4]								EEPROM
0x00A	ZERO_POS[3:0]				Z_EDGE	Z_PUL_WID[2:0]			EEPROM

(1) Auto Setting by SPI with 'Auto Setting Zero-Position' Command

This method can only change the zero-position to current mechanical position.

(2) Manual Calculate Zero Degree and Write Register by SPI

- (a) Write data 0x000 to register 'ZERO_POS[11:0]' by SPI;
- (b) Read out the angle data of current position by SPI;
- (c) Calculate the target Zero-Position value, write it to 'ZERO_POS[11:0]' by SPI;

For both auto and manual methods, an extra SPI 'Program EEPROM' operation is necessary for storing the new zero-position value into EEPROM, and at least to wait 6 seconds for the EEPROM programming without any other operation.

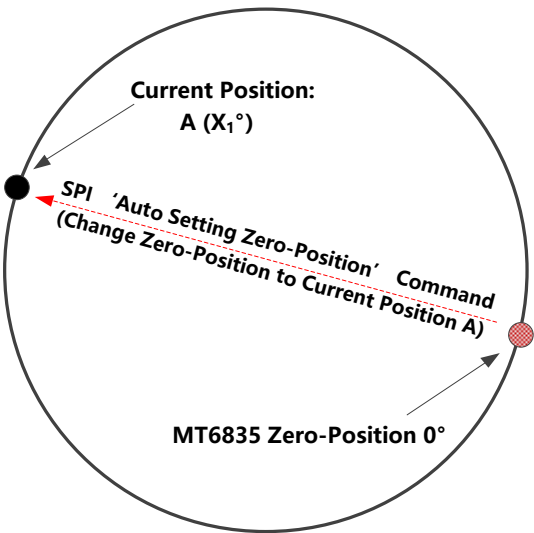


Figure 26-1: Auto Setting

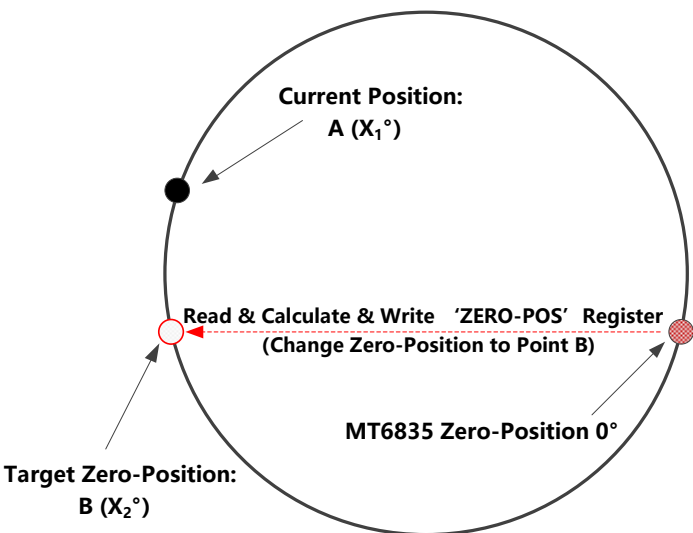


Figure 26-2: Manual Change

9. Calibration

Three calibration levels are designed in MT6835: the first level of Factory Calibration is done by MagnTek before the chip is delivered to the user; The second level of User Auto-Calibration could be optionally done by the user at a constant rotation speed; The third level User NLC Calibration could be optionally done by the user with SPI interface and external precision angle reference such as a precision optical encoder.

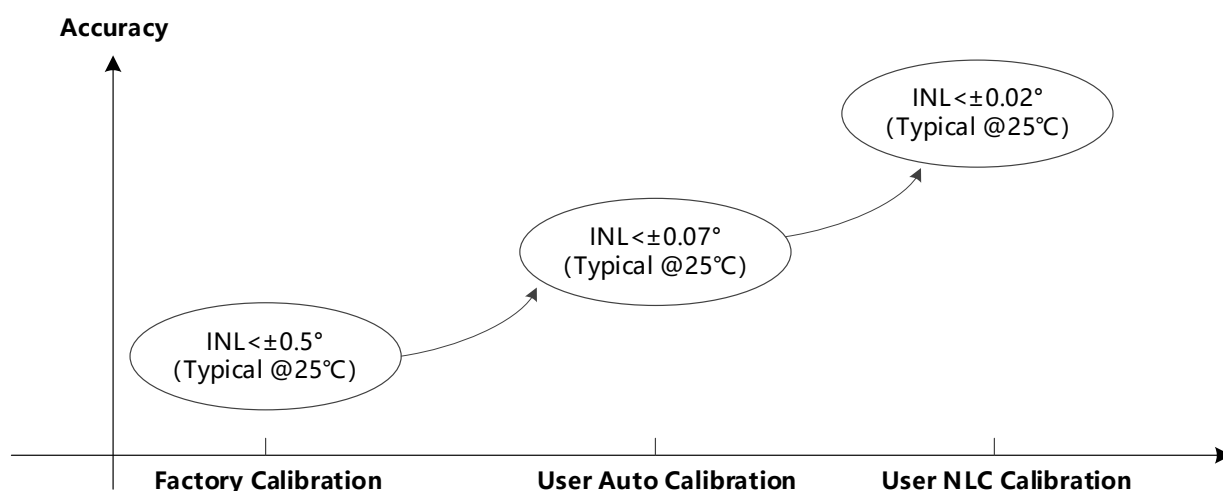


Figure 27: Three Calibration Levels

9.1 Factory Calibration

MT6835 is factory calibrated before it is delivered to the user. The original offset, gain-mismatch and phase-error of sine/cosine signals are calibrated by FT testing as shown in Figure 28, thus the INL could be reduced to less than $\pm 0.5^\circ$ (typical case).

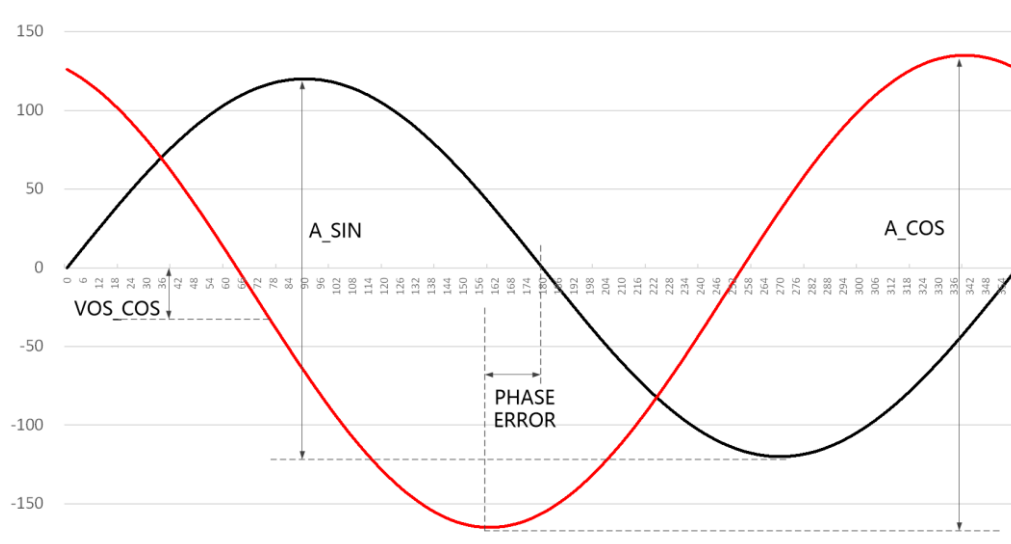


Figure 28: Offset, Gain-Mismatch and Phase-Error

21-Bit High Accuracy Magnetic Angle Encoder IC

9.2 User Auto-Calibration

User Auto-Calibration allows the user to configure MT6835 to fit with a specific magnetic field and assembling condition in an application system. Let the motor rotate at a constant speed and pull up Pin.4 (CAL_EN) to VDD will enable this calibration, then the DSP of MT6835 will automatically calculate compensation coefficients compensate the calibration process. Typically, INL will be reduced to less than $\pm 0.07^\circ$ when User Auto-Calibration has been done successfully.

The operation steps:

- (1) Setting the rotation speed register for User Auto-Calibration;

User Auto-Calibration Rotation Speed Register (EEPROM)

Reg. AUTOCAL_FREQ[2:0]	Description (Rotation Speed: RPM)
0x0	$3200 \leq \text{Speed} < 6400$
0x1	$1600 \leq \text{Speed} < 3200$
0x2	$800 \leq \text{Speed} < 1600$
0x3	$400 \leq \text{Speed} < 800$
0x4	$200 \leq \text{Speed} < 400$
0x5	$100 \leq \text{Speed} < 200$
0x6	$50 \leq \text{Speed} < 100$
0x7	$25 \leq \text{Speed} < 50$

- (2) Rotating the system at a constant speed with the present range;
- (3) Pull up CAL_EN (Pin.4) to VDD to start User Auto-Calibration. Keep the system rotating at the set speed for more than 64 rounds. Monitor PWM (Pin.10) Output (When Pin.4 tied to VDD, Pin.10 is also configured as calibration status output).

PWM Output Duty-Cycle	Description
50%	Running Auto Calibration
25%	Calibration Failed
>99%	Calibration Successful

The calibration status also could be read from register 0x113[7:6]:

0x113[7:6]=00, No Calibration ; 0x113[7:6]=01, Running Calibration;
 0x113[7:6]=10, Calibration Failed; 0x113[7:6]=11, Calibration Successful;

- (4) If calibration has failed, please check the system and then re-calibration again;
- (5) When calibration is successful, please wait >6s and power-down the chip;
- (6) Other operations cannot be performed until the chip is powered on again;

9.3 User NLC Calibration

The User NLC (Non-Linearity Compensation) Calibration of MT6835 provides a 256-point look-up table for further INL improvement as shown in Figure 29. A possible calibration system is shown in Figure 30, a high precision optical encoder (≥ 21 -bit) is needed to provide the ideal angle reference when doing the User NLC Calibration.

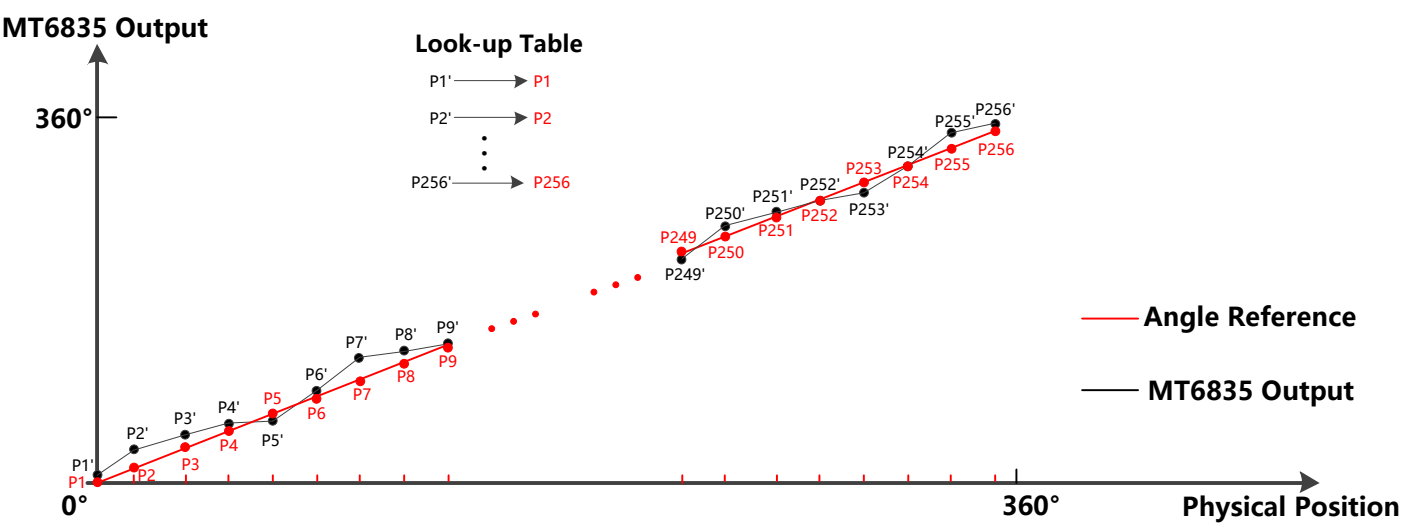


Figure 29: User NLC Calibration

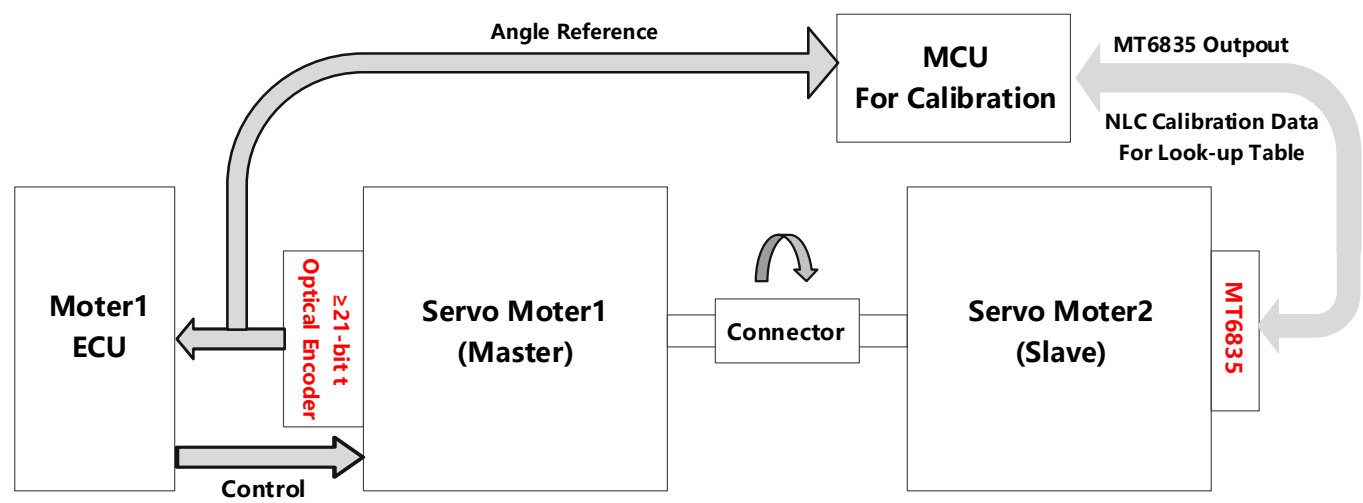


Figure 30: Possible User NLC Calibration System

21-Bit High Accuracy Magnetic Angle Encoder IC

10. Register Map

Address	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0	Type
0x001	USER_ID[7:0]								EEPROM
0x002	Not Used								NA
0x003	ANGLE[20:13]								Read Only
0x004	ANGLE[12:5]								Read Only
0x005	ANGLE[4:0]					STATUS[2:0]			Read Only
0x006	CRC[7:0]								Read Only
0x007	ABZ_RES[13:6]								EEPROM
0x008	ABZ_RES[5:0]						ABZ_OFF	AB_SWAP	EEPROM
0x009	ZERO_POS[11:4]								EEPROM
0x00A	ZERO_POS[3:0]					Z_EDGE	Z_PUL_WID[2:0]		EEPROM
0x00B	Z_PHASE[1:0]		UVW_MUX	UVW_OFF	UVW_RES[3:0]				EEPROM
0x00C	MagnTek Use Only		NLC_EN	PWM_FQ	PWM_POL	PWM_SEL[2:0]			EEPROM
0x00D	MagnTek Use Only					ROT_DIR	HYST[2:0]		EEPROM
0x00E	GPIO_DS	AUTOCAL_FREQ[2:0]			MagnTek Use Only				EEPROM
0x00F	MagnTek Use Only								EEPROM
0x010	MagnTek Use Only								EEPROM
0x011	MagnTek Use Only					BW[2:0]			EEPROM
0x012	MagnTek Use Only								EEPROM
0x013 ~ 0x0D2	NLC Byte 0x00~0xBF (Total 192 Bytes)								EEPROM

Warning: Do Not Change the 'MagnTek Use Only' Bits

(1) 0x001 User_ID (EEPROM)

Address	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0	Type
0x001	USER_ID[7:0]								EEPROM

- 'USER_ID[7:0]' is a free byte for the user.

21-Bit High Accuracy Magnetic Angle Encoder IC

(2) 0x003~ 0x006 Angle Data Register (Read Only)

Address	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0	Type
0x003	ANGLE[20:13]								Read Only
0x004	ANGLE[12:5]								Read Only
0x005	ANGLE[4:0]					STATUS[2:0]			Read Only
0x006	CRC[7:0]								Read Only

- 0~360° absolute angle θ could be calculated by the below formula with ANGLE[20:0]:

$$\theta = \frac{\sum_{i=0}^{20} \text{ANGLE}[i] \cdot 2^i}{2^{21}} \cdot 360^\circ$$

- 'STATUS[2:0]' indicates some warnings of the chip;

STATUS[2:0]	Notes
Bit [0]	Rotation Over Speed Warning; Default '0' , Logic '1' for Warning
Bit [1]	Weak Magnetic Field Warning; Default '0' , Logic '1' for Warning
Bit [2]	Under Voltage Warning; Default '0' , Logic '1' for Warning

- 'CRC[7:0]' gives the redundant check of bits ANGLE[20:0] and STATUS[2:0], totally 24bits. The CRC polynomial: X^8+X^2+X+1 , MSB (ANGLE[20]) shifts in first.

(3) 0x007~ 0x008 ABZ Resolution and Related Registers (EEPROM)

Address	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0x007	ABZ_RES[13:6]							
0x008	ABZ_RES[5:0]				ABZ_OFF		AB_SWAP	

- 'ABZ_RES[13:0]' configures the resolution of AB output, please refer to Chapter 7.3;

Reg. ABZ_RES[13:0]	AB Resolution (Pulse per. Round)	Factory Default Setting
0x0000	1	√
0x0001	2	
... ..		
0x3FFE	16,383	
0x3FFF	16,384	

21-Bit High Accuracy Magnetic Angle Encoder IC

- 'ABZ_OFF' configures the on/off state of ABZ output;

Reg. ABZ_OFF	ABZ Output	Factory Default Setting
0x0	ON	√
0x1	OFF	

- 'AB_SWAP' configures the swapping of incremental output A and B;

Reg. AB_SWAP	AB Output	Factory Default Setting
0x0	No Swap	√
0x1	Swap	

(4) 0x009~ 0x00A Z Pulse Related Registers (EEPROM)

Address	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0x009	ZERO_POS[11:4]							
0x00A	ZERO_POS[3:0]				Z_EDGE	Z_PUL_WID[2:0]		

- 'ZERO_POS[11:0]' configures the Zero-Position of MT6835, it is effective for all outputs;

Reg. ZERO_POS[11:0]	Absolute Position (°)	Factory Default Setting
0x000	0	√
0x001	0.088	
... ..		
0xFFE	359.824	
0xFFF	359.912	

- 'Z_EDGE' configures the relationship of Z pulse edge and zero degree;

Reg. Z_EDGE	Z Pulse Edge with 0°	Factory Default Setting
0x0	Rising Edge Aligned to 0°	√
0x1	Falling Edge Aligned to 0°	

21-Bit High Accuracy Magnetic Angle Encoder IC

- 'Z_PUL_WID[2:0]' configures the width of Z pulse.

Reg. Z_PUL_WID[2:0]	Width (LSBs/°)	Factory Default Setting
0x0	1	√
0x1	2	
0x2	4	
0x3	8	
0x4	16	
0x5	60°	
0x6	120°	
0x7	180°	

(5) 0x00B Z_PHASE and UVW Related Registers (EEPROM)

Address	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0x00B	Z_PHASE[1:0]		UVW_MUX	UVW_OFF	UVW_RES[3:0]			

- 'Z_PHASE[1:0]' configures the phase between Z and A/B output, please refer to Figure 12 on Page-15;
- 'UVW_MUX' configures UVW pins (Pin.1~Pin.3) output UVW or -A-B-Z;

Reg. UVW_MUX	UVW Output Pin	Factory Default Setting
0x0	UVW	√
0x1	-A-B-Z	

- 'UVW_OFF' configures the on/off state of the UVW output;

Reg. UVW_OFF	UVW Output	Factory Default Setting
0x0	ON	√
0x1	OFF	

- 'UVW_RES[3:0]' configures the resolution of UVW output;

Reg. UVW_RES[3:0]	UVW Pole Pairs	Factory Default Setting
0x0	1	√
0x1	2	
... ..		
0xE	15	
0xF	16	

21-Bit High Accuracy Magnetic Angle Encoder IC

(6) 0x00C PWM and NLC Calibration Related Registers (EEPROM)

Address	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0x00C	MagnTek Use Only		NLC_EN	PWM_FQ	PWM_POL	PWM_SEL[2:0]		

- 'NLC_EN' enables NLC Calibration

Reg. NLC_EN	NLC Calibration	Factory Default Setting
0x0	Disabled	√
0x1	Enabled	

- 'PWM_FQ' configures the PWM frame frequency ;

Reg. PWM_FQ	PWM Frame Frequency	Factory Default Setting
0x0	994 Hz	√
0x1	497 Hz	

- 'PWM_POL' configures PWM effective voltage level;

Reg. PWM_POL	PWM Polarity	Factory Default Setting
0x0	High Voltage Effective	√
0x1	Low Voltage Effective	

- 'PWM_SEL[2:0]' configures the PWM output source;

Reg. PWM_SEL[2:0]	PWM Data Source	Factory Default Setting
0x0	12-bit Angle Data	√
0x2	12-bit Velocity Data	
Others	Factory Test Data	

(7) 0x00D (EEPROM)

Address	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0x00D	MagnTek Use Only				ROT_DIR	HYST[2:0]		

- 'ROT_DIR' configures the rotation direction (logic 0 for CCW), please refer to Figure 7;

Reg. ROT_DIR	Rotation Direction	Factory Default Setting
0x0	Counter-Clockwise	√
0x1	Clockwise	

21-Bit High Accuracy Magnetic Angle Encoder IC

- 'HYST[2:0]' configures the hysteresis window for angle output

Reg. HYST[2:0]	Hysteresis Window	Factory Default Setting
0x0	0.022°	
0x1	0.044°	
0x2	0.088°	
0x3	0.176°	
0x4	0	
0x5	0.003°	
0x6	0.006°	
0x7	0.011°	√

(8) 0x00E (EEPROM)

Address	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0x00E	GPIO_DS	AUTOCAL_FREQ[2:0]			MagnTek Use Only			

- 'GPIO_DS' configures the GPIO drive capability;

Reg. GPIO_DS	GPIO Drive Capability	Factory Default Setting
0x0	Default Level	√
0x1	Doubled Level	

- 'AUTOCAL_FREQ[2:0]' configures the rotation speed for Auto Calibration;

Reg. AUTO_CAL_FREQ[2:0]	Description (Rotation Speed: RPM)	Factory Default Setting
0x0	3200≤Speed<6400	
0x1	1600≤Speed<3200	
0x2	800≤Speed<1600	
0x3	400 ≤Speed< 800	√
0x4	200≤Speed<400	
0x5	100≤Speed<200	
0x6	50≤Speed<100	
0x7	25≤Speed<50	

21-Bit High Accuracy Magnetic Angle Encoder IC

(8) 0x011 System Bandwidth (EEPROM)

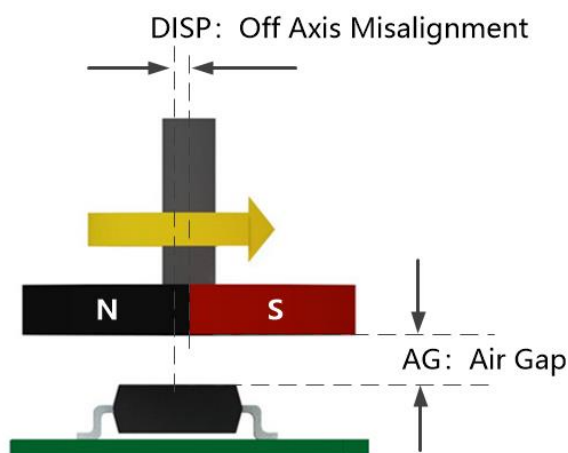
Address	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0x011	MagnTek Use Only					BW [2:0]		

- 'BW[2:0]' configures the MT6835's system bandwidth.

Reg. BW[2:0]	Description	Response	Factory Default Setting
0x0	System Bandwidth Baseline	Slow (Better Noise)	
0x1	Baseline x 2		
0x2	Baseline x 4		
0x3	Baseline x 8		
0x4	Baseline x 16		
0x5	Baseline x 32		√
0x6	Baseline x 64		
0x7	Baseline x 128	Fast (Worse Noise)	

11. Magnet Placement

It is required that the magnet's center axis be aligned with the sensing element center of MT6835 with the air-gap as small as possible. Any misalignment introduces additional angle error and big air-gap also weakens the magnetic field which could be sensed by the device. Magnets with larger diameter are more tolerant to DISP (off-axis misalignment) and big AG (air-gap between Magnet and device).



INL vs. DISP for $\Phi 10$ magnet

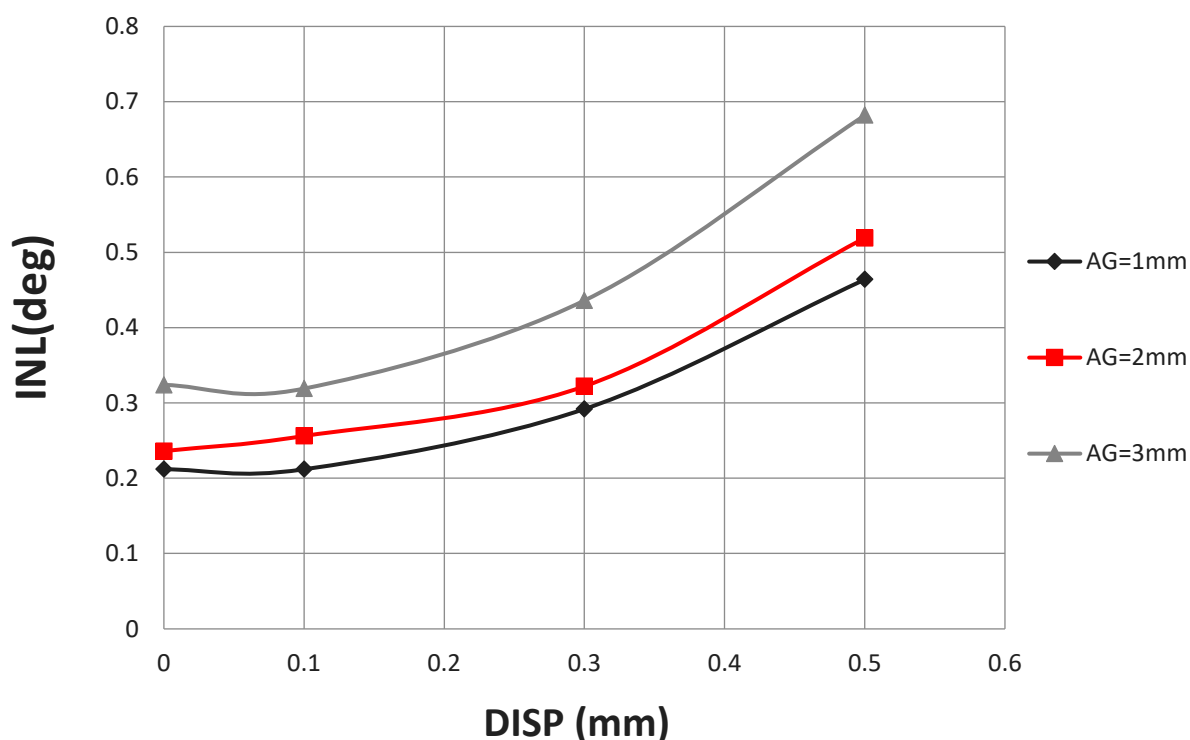
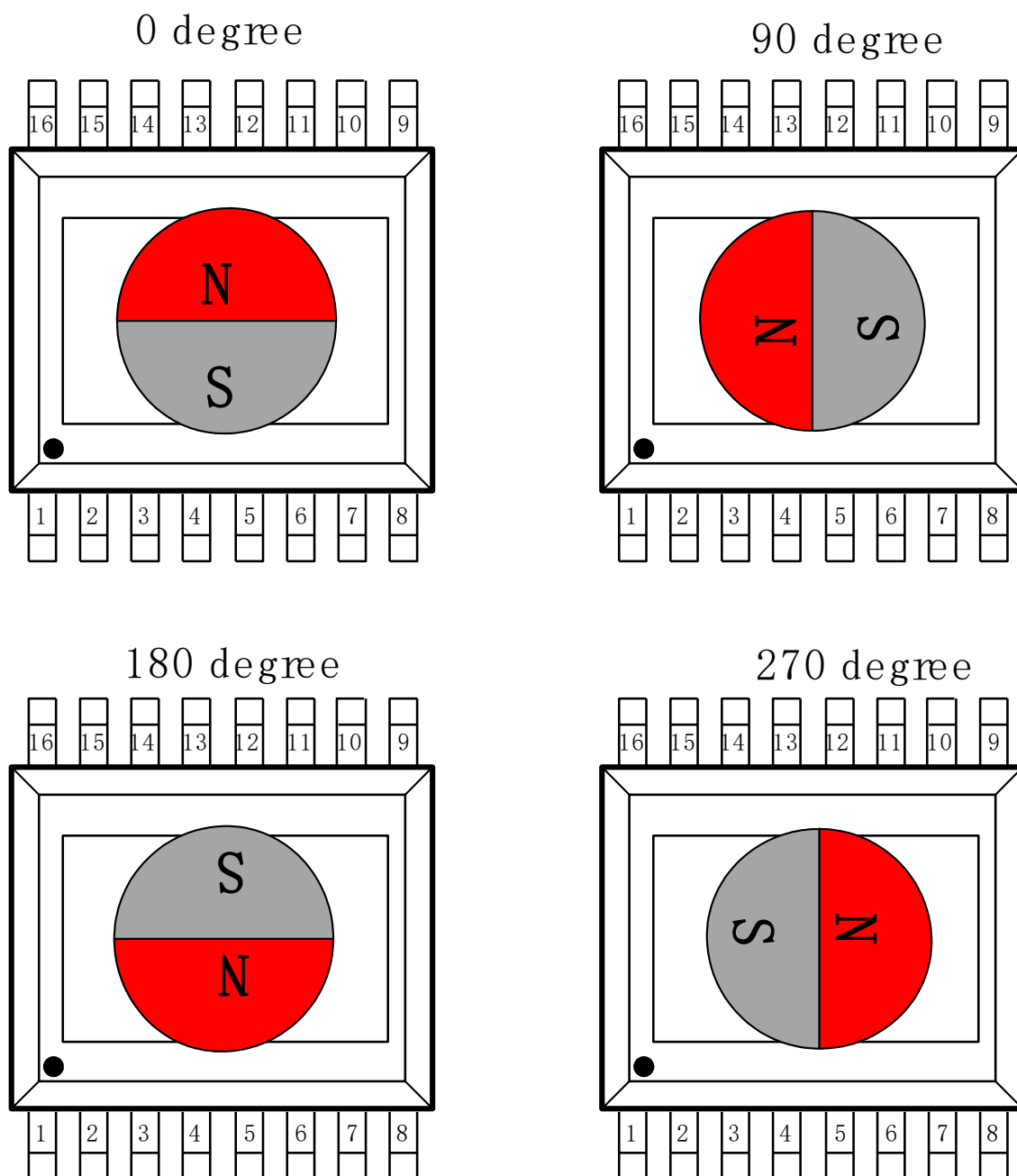


Figure 31: INL vs DISP and AG with Factory Calibration Only

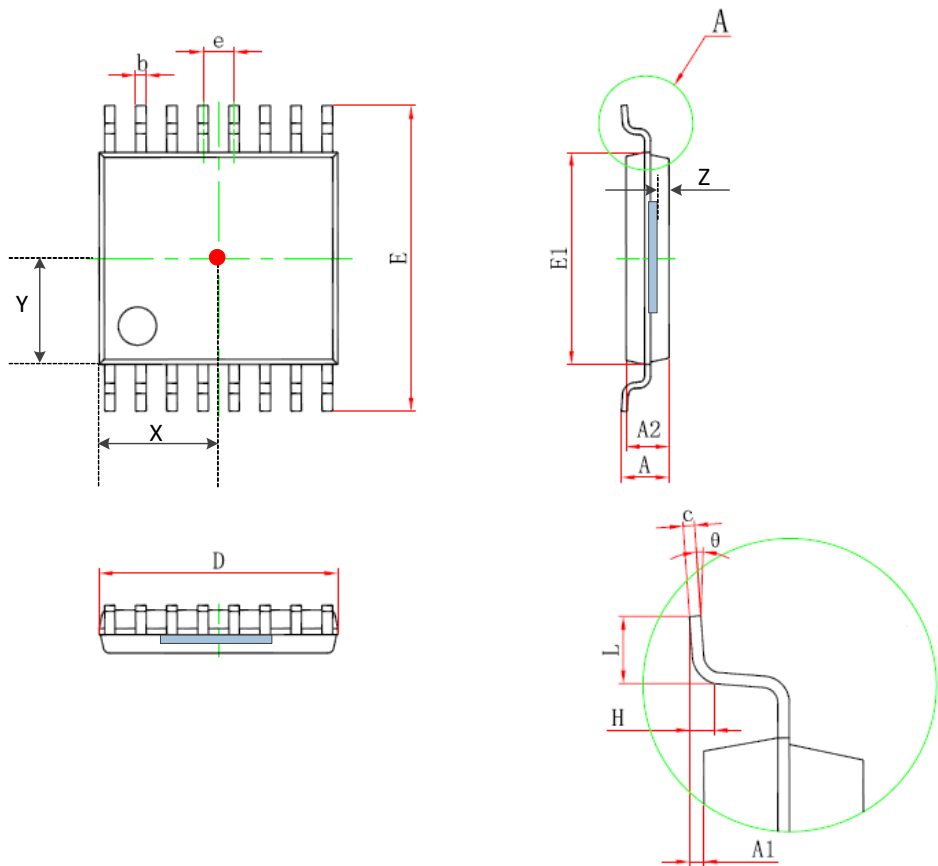
12. Mechanical Angle Direction

Top View



Counter-Clockwise Rotation (CCW)
(with Register 'ROT-DIR' =0)

13. Package Information



Symbol	Dimensions in Millimeters		Dimensions in Inches	
	Min.	Max.	Min.	Max.
D	4.900	5.100	0.193	0.201
E	6.250	6.550	0.246	0.258
b	0.190	0.300	0.007	0.012
c	0.090	0.200	0.004	0.008
E1	4.300	4.500	0.169	0.177
A		1.200		0.047
A2	0.800	1.000	0.031	0.039
A1	0.050	0.150	0.002	0.006
e	0.65 (BSC)		0.026 (BSC)	
L	0.500	0.700	0.020	0.028
H	0.25 (TYP)		0.01 (TYP)	
θ	1°	7°	1°	7°
X	2.450	2.550	0.097	0.101
Y	2.150	2.250	0.085	0.089
Z	0.210	0.370	0.016	0.024

14. Copy Rights and Disclaimer

1. This document may not be reproduced or duplicated, in any form, in whole or in part without prior written consent of MagnTek . Copyrights © 2021, MagnTek Incorporated.
2. MagnTek reserves the right to make changes to the information published in this document at anytime without notice.
3. MagnTek's products are limited for use in normal commercial applications. MagnTek's products are not to be used in any device or system, including but not limited to medical life support equipment and system.

For the latest version of this document, please visit our website: www.magntek.com.cn

15. Revision History

Revision	Date	Comments
0.5	2021.01	Initial Release as Draft
1.0	2021.04	Formal Release
1.1	2022.02	Update Figure 24 Timing Add AB Maximum Frequency
1.2	2022.05	Update Chapter 7.6.7 & 7.6.8
1.3	2022.12	Update Figure 7 Update Default Hyst=0.011° Update Chapter 7.6.2